A Stepwise Approach for Monitoring Spring Maize Phenology Using Fused Multisource Remote Sensing Data

Yuyang Ma, Haixiang Guan, Zhou Fang, Yan Shuai, Bo Huang, Xuecao Li, and Jianxi Huang, *Senior Member, IEEE*

***Abstract*—Accurate crop phenology information is essential for optimizing agricultural management and understanding agroecosystem responses to climate change. However, existing datasets are usually constrained by limited spatial and temporal resolutions and by oversimplified phenological stages. In this study, we proposed a stepwise strategy for fine-scale crop phenology monitoring by integrating multisource remote sensing data. First, we generated dense time-series remote sensing data at 30-m spatial resolution using a quick spatiotemporal fusion (STF) approach with coarse- and fine-resolution scale transformation errors (STEs) and a pixel-based synthesis base image pair (STEPSBI), enhancing temporal continuity while preserving spatial detail. Then, we assessed the separability of six phenological stages, based on the Gaussian probability density distributions of candidate features, which enabled the selection of optimal input combinations. Finally, we utilized random forest (RF) regression to model vegetative and reproductive growth stages, generating the spring maize phenology from 2001 to 2020 in Northeast China. Results indicated that the proposed stepwise strategy outperformed 1-D U-Net, phenology Seq2Seq network (PSeqNet), and improved shape model fitting models, with root-mean-square error (RMSE) ranging from 5.29 to 9.45 days in six phenological stages. Furthermore, compared with models using only spectral reflectance, vegetation indices, or temperature-derived factors, our approach reduced total RMSE by 15.70, 20.33, and 23.13 days, respectively. Moreover, the STEPSBI-fused images exhibited high agreement with the raw imagery, with the correlation coefficients (R) of the bands ranging from 0.81 to 0.88, effectively overcoming the spatial and temporal resolution limitations of remote sensing images. The generated dataset is consistent with existing phenology products from Landsat (30 m), Moderate Resolution Imaging Spectroradiometer (MODIS) (500 m), and GLASS (1 km), while providing finer spatial details and capturing interannual variability. These advantages support precision agricultural management and improve our understanding of the agroecosystem response to regional climate change.**

***Index Terms*—Long-term span, maize phenology, regional scale, spatiotemporal fusion (STF), stepwise strategy.**

I. Introduction

CROP phenology refers to the periodic progression of plants from emergence, leaf expansion, and flowering to maturity, influenced by genetic, meteorological, and soil conditions. Multistage phenology provides critical information for agricultural management, enabling precise decisions in irrigation scheduling, targeted fertilization, pest control, and harvesting [1], [2], [3]. Moreover, shifts in crop phenology can significantly impact biogeochemical and biogeophysical processes in agroecosystems, including photosynthesis, evapotranspiration, and surface energy fluxes, which in turn influence carbon sequestration and water cycling [4], [5], [6]. Therefore, acquiring fine-scale crop phenology information across large spatial scales, long-term spans, and multiple growth stages is crucial for advancing precision agriculture and improving our understanding of agricultural system responses to climate change.

With the development of various sensors and satellites, remote sensing imagery has been widely applied to crop phenology monitoring [7]. Optical temporal remote sensing images are the primary data sources for large-scale crop phenology monitoring, encompassing the Moderate Resolution Imaging Spectroradiometer (MODIS) [8], [9], Advanced Very High Resolution Radiometer (AVHRR) [10], Landsat, and Sentinel-2 imagery [11], [12]. MODIS and AVHRR offer high temporal resolution, facilitating the detection of multistage phenology from sowing to harvest. However, their coarse spatial resolution (500–1 km) limits the capture of phenological information at the field scale. In contrast, Landsat imagery has higher spatial resolution (30 m) and a longer temporal span, making it widely utilized for monitoring land surface dynamics. However, gathering adequate medium spatial resolution remote sensing images to identify multistage phenology is hindered by cloud contamination and the 16-day revisit interval [12]. Therefore, in fine-scale crop phenology monitoring, there is a tradeoff between spatial resolution and phenological stage diversity. Specifically, phenological datasets with high spatial resolution typically cover fewer

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Yuyang Ma is with the College of Geography and Planning, Chengdu University of Technology, Chengdu 610059, China (e-mail: yuyangma@cdut.edu.cn).

Haixiang Guan, Zhou Fang, Bo Huang, and Jianxi Huang are with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu 610000, China (e-mail: haixiangguan@cau.edu.cn; 2020114112@my.swjtu.edu.cn; hb@my.swjtu.edu.cn; jxhuang@cau.edu.cn).

Yan Shuai is with Shandong Rural Remote Sensing Application Center, Shandong Academy of Agricultural Sciences, Jinan 250100, China (e-mail: yanshuaicau@126.com).

Xuecao Li is with the College of Land Science and Technology, China Agricultural University, Beijing 100083, China (e-mail: xuecaoli@cau.edu.cn).

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stages, whereas those capturing multiple stages often lack adequate spatial detail. Additionally, the spectral differences of crops across successive phenological stages are often subtle, and external uncertainties arising from long-term climatic variability further complicate multistage phenology identification. Spatiotemporal fusion (STF) effectively addresses the issue of insufficient spatiotemporal resolution in remote sensing imagery, leveraging temporal variation trends and spatial relationships between high temporal-coarse spatial resolution and low temporal-fine spatial resolution data. Currently, various STF algorithms have been developed to fuse MODIS and Landsat data, generating high-resolution images for phenological monitoring [13], [14], [15], [16], [17], [18], [19], [20]. For instance, Nietupski et al. [21] fused the normalized difference vegetation index (NDVI) time series using the enhanced spatial and temporal adaptive reflectance fusion model (ESTARFM) method. Subsequently, they detected the start of season (SOS), end of season (EOS), and dormancy using a double-logistic approach. Sisheber et al. [22] utilized ESTARFM to fuse EVI2 time series and employed a dynamic threshold to track crop phenology dynamics.

Furthermore, Sisheber et al. [23] proposed a knowledge-driven ESTARFM method to fuse enhanced vegetation index (EVI) time series and identify SOS and EOS in heterogeneous agricultural landscapes. However, two limitations remain: 1) ESTARFM has stringent requirements on input images, necessitating two pairs of cloudless images from before and after the predicted time, which restricts the fusion during the crop growing season due to continuous cloud occlusion and 2) the knowledge-driven ESTARFM approach depends on prior information, such as long-term records of crop sowing and harvesting, which are laborious and difficult to implement at the regional scales. To address these challenges, our previous work developed a quick STF incorporating coarse- and fine-resolution scale transformation errors (STEs) and a pixel-based synthesis of base image pair (STEPSBI) [19], aiming to restore the spatial details in remote sensing images under continuous cloud occlusion. Specifically, we developed a pixel-based optimum image synthesis approach based on the "similarity" and "consistency" criteria, mitigating the missing of base images caused by continuous cloud occlusion. Second, the method quantified the temporal variation residuals from the reference to prediction time, as well as the spatial STEs between coarse- and fine-resolution data. Super-pixel segmentation results from high-resolution images are subsequently used as spatial weights to correct these errors, enhancing the spatial continuity and detail fidelity of the fused images. Additionally, this method is implemented on the GEE platform, offering higher fusion efficiency to support large-scale phenological monitoring. However, there is a lack of large-scale applications in the agricultural domain.

Conventional phenological detection approaches, such as threshold-based and curve-fitting methods, can effectively capture long-term phenological patterns at the regional scale [7], [12], [27]. Nonetheless, three main limitations exist: 1) thresholds or curve shapes need to be predefined artificially; 2) these methods primarily distinguish the emergence and maturity date (MD), making it difficult to divide multiple intermediate stages; and 3) they rely on postseason monitoring based on time-series curves. In contrast, the random forest (RF) method does not require a predefined fitting function, evaluates multiple feature dimensions simultaneously, and is robust to temporal fluctuations in individual variables. Furthermore, training the RF regression model enables phenological identification at various stages, including the three-leaf date (V3), seven-leaf date (V7), jointing date (JD), tassel date (TD), milky date (MID), and MD. This approach improves the precision of phenological distinction under complex environments and realizes the shift from postseason monitoring to within-season detection.

Although satellite remote sensing and STF techniques have advanced, monitoring multistage crop phenology at the regional scale remains challenging. This study focuses on the scientific problem of bridging the scale gap between ground observations and satellite monitoring, aiming to achieve fine-scale phenological mapping that is multistage, large-scale, and long-term. There are two technological challenges: 1) at the regional scale, acquiring dense time-series imagery throughout the growing season is difficult and 2) over the long-term spans, remote sensing-based phenology studies have often focused on one or two key stages, with limited capability to capture the continuous progression of multiple phenological stages. These limitations could arise from subtle spectral differences across various stages, as well as the uncertainties introduced by prolonged environmental changes. Addressing these challenges is expected to enhance the accuracy and comprehensiveness of regional phenology monitoring, improve understanding of crop-climate interactions, and provide critical data support for agricultural management and food security.

To sum up, we developed a stepwise strategy for monitoring six stages of crop phenology in Northeast China at 30-m resolution from 2001 to 2020. The approach aims to accurately identify the key within-season and historical phenological stages and to capture their spatial heterogeneity at the field level. The remainder of this article is organized as follows. Section II describes the study area and data. Section III outlines the proposed approach for fine-scale crop phenology monitoring. Section IV analyzes the results. Section V provides the discussions. Section VI concludes this study.

II. Materials

A. Study Area

The study area is located in northeastern China, covering Heilongjiang, Jilin, and Liaoning provinces between 38.62°N and 53.43°N latitude and 118.83°E and 135.10°E longitude (Fig. 1). This region experiences a cold-temperate continental semihumid climate, with annual precipitation ranging from 500 to 800 mm, approximately 60% of which occurs between July and September. The average temperature during the growing season exceeds 17 °C, and the frost-free period lasts 140–170 days. Renowned for spring maize production, this area accounts for 31.51% of China's maize planting area and roughly 34.33% of national production (http://www.stats.gov.cn/sj/ndsj/, accessed on November 10, 2024). The growing season typically spans from May to October, and

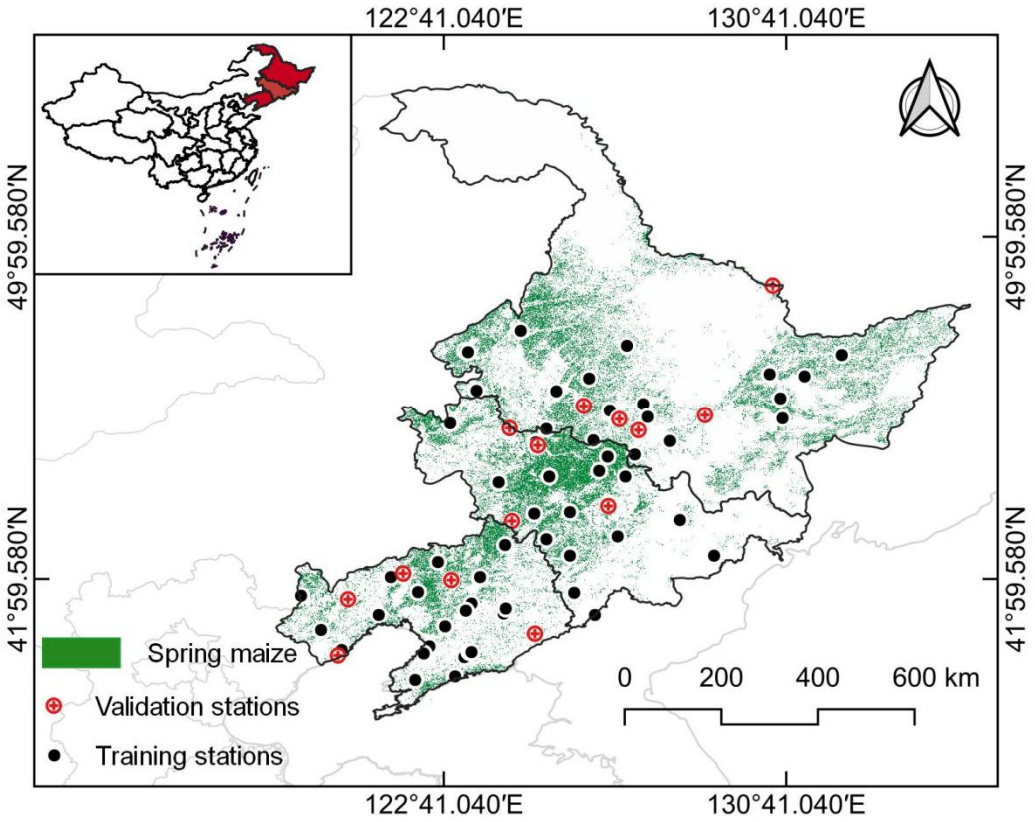


Fig. 1. Spatial distribution of spring maize, training stations, and validation stations in Northeast China.

TABLE I
SIX PHENOLOGICAL STAGES AND DATE OF SPRING MAIZE

Phenological stage	Description
Three-leaf stage	The date when the third leaf is fully expanded.
Seven-leaf date	The date when the seventh leaf is fully expanded.
Jointing date	The date when the internodes at the base of the plant stem elongate.
Tassel date	The date when the main tassel of the plant emerges 3-5 cm above the top leaf.
Milky date	The date when the kernel color in the middle of the plant ear begins to show the inherent color of the variety, and the endosperm turns milky to mushy.
Maturity date	The date when the dry weight of maize grains first reaches the maximum.

phenological variations are influenced by climate warming, soil degradation, and varietal changes [24], [25], [26].

B. Data Acquisition and Preprocessing

This study acquired ground phenology observation data to train and validate the phenological model. Then, we utilized Landsat image, fused Landsat and MODIS, climate reanalysis, and maize spatial distribution data to develop a stepwise strategy for detecting multistage maize phenology, generating a dataset, and analyzing its spatiotemporal changes.

1) Ground Phenology Observation Data: This study obtained ground observation data from the Agricultural Meteorological Stations (AMSs) of China Meteorological Administration (CMA) (https://data.cma.cn/). The data were collected by trained agricultural professionals and ensured quality through China Agricultural Weather Monitoring System [27]. A phenological date was recorded when more than half of the maize fields reached the corresponding stage. The phenological stages included V3, V7, JD, TD, MID, and MD of maize (Table I).

2) Remote Sensing Images: This study utilized Landsat surface reflectance data for spring maize phenology mapping from 2001 to 2020, including Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+), and Operational Land Imager ETM (OLI). The data were corrected for radiometric and topographic effects, with atmospheric correction performed using the Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS) [28], and clouds and shadows were removed using the Fmask algorithm [29]. Linear regression was also applied to align TM and reflectance values with OLI, mitigating biases from sensor and data quality differences [30].

Monitoring the six phenological stages requires high temporal resolution because the stages occur at one–two week

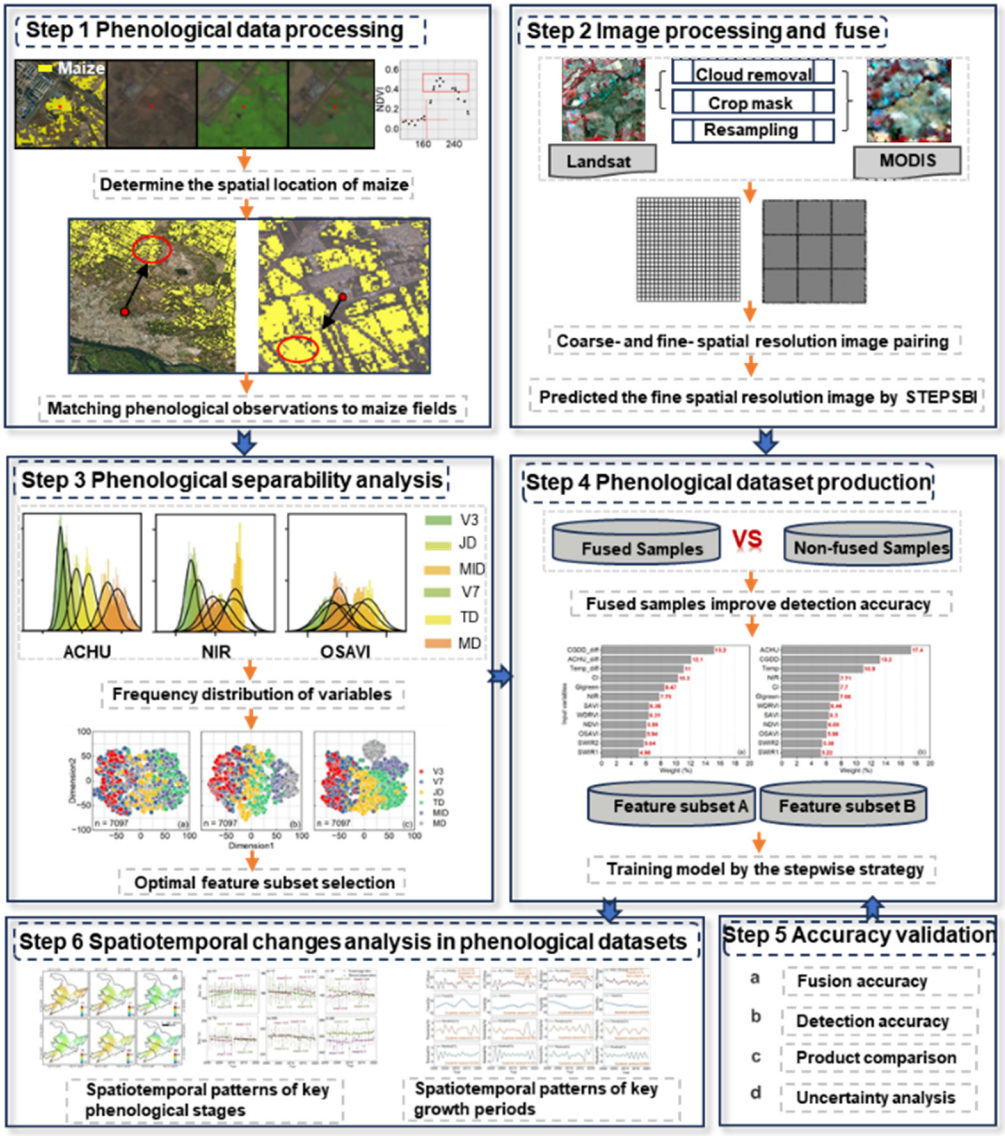


Fig. 2. Overall framework for mapping phenology dataset using the stepwise strategy.

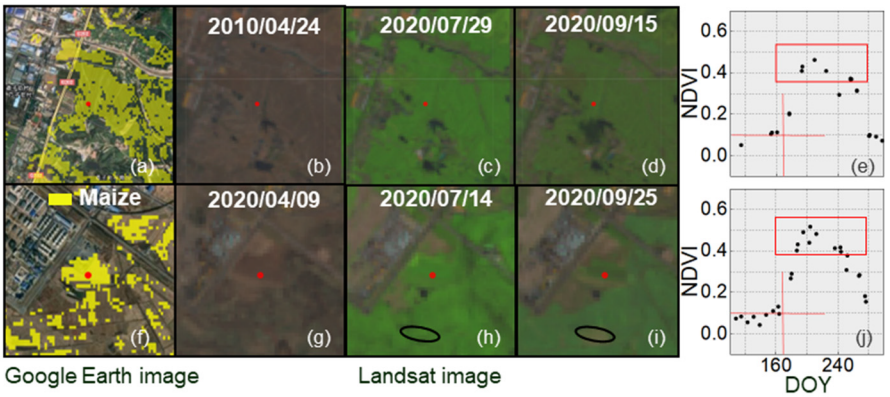


Fig. 3. Flowchart for determining the location of maize samples. (a) and (f) Spatial location of the true-color composite Google image and the maize layer, respectively. (b)–(d) and (g)–(i) True-color composite images of Landsat TM and OLI. (e) and (j) Reflect the NDVI time series at the corresponding locations, where the rapid rise near the JD (red cross lines) and the slight decrease after pollination (red rectangles) are highlighted.

intervals, which makes it difficult for Landsat images to comprehensively cover all maize planting areas. Thus, this study utilized all available MOD09A1 reflectance images from 2001 to 2020, with an eight-day temporal resolution

TABLE II
REMOTE SENSING FEATURES AND THEIR FORMULAS IN THIS STUDY

Abbr	Name	Equation	Reference
NDVI	Normalized Difference Vegetation Index	$\frac{NIR - RED}{NIR + RED}$	[32]
SAVI	Soil-Adjusted Vegetation Index	$\left[\frac{NIR - RED}{NIR + RED + L}\right] \times (1 + L)$	[33]
OSAVI	Optimized soil-adjusted vegetation index	$\frac{1.5 \times (NIR - RED)}{NIR + RED + 0.16}$	[36]
GIgreen	Green Chlorophyll Vegetation Index	$\frac{NIR}{GREEN} - 1$	[37]
WDRVI	Wide Dynamic Range Vegetation Index	$\frac{0.1 \times NIR - RED}{0.1 \times NIR + RED}$	[38]
CI	Clay Index	$\frac{SWIR1}{SWIR2}$	[39]

and 500-m spatial resolution. We removed clouds and cloud shadows using the StateQA band and filled gaps caused by cloudy pixels through linear interpolation [31]. Finally, MOD09A1 images were resampled to 480 m to match the spatial resolution of Landsat images in integer multiples.

Twenty-one spectral indices and six-band reflectance were extracted for their potential correlation with crop physiology [7] (Fig. S1 and Table II). Among them, NDVI is widely used due to its sensitivity to chlorophyll and strong correlation with canopy photosynthesis [32]. Soil-adjusted vegetation index (SAVI) reduces the effects of soil brightness and aerosols while maintaining sensitivity to green vegetation [33], [34], [35]. Optimized SAVI (OSAVI) effectively supports early-stage crop health diagnostics by removing soil background influence [36]. We also included GIgreen, which has been shown to capture key phenological stages [37]. Wide dynamic range vegetation index (WDRVI) correlates linearly with maize leaf area index, making it sensitive to biomass variations [38]. Clay index (CI) reflects soil conditions and soil nutrient content that affect crop growth [39]. Additionally, spectral bands in the near infrared response (NIR), SWIR1, and SWIR2 are sensitive to leaf structure, dry matter, and leaf water content [2]. We present probability density distributions of these variables across various phenological stages in Fig. S1, with a more detailed discussion in Section III-C.

3) Meteorological Data: This study acquired daily surface solar radiation (ShortWave), total precipitation (Pre), and volumetric soil water at layer 2 (Soilwater, 7–28 cm) from the enhanced ERA5_land daily dataset. We obtained hourly air temperature at 2 m from ERA5_land hourly, with a spatial resolution of 0.1° × 0.1° [40]. They were resampled to 30 m to match the spatial resolution of Landsat and better align with maize fields. We then aggregated the hourly temperature data into daily mean (Temp), maximum (Temp_max), and minimum (Temp_min) temperatures. Using these temperature variables, we calculated accumulated heat unit (ACHUs) and cumulative growth degree days. Cumulative growing degree day (CGDD) represents the accumulated effective temperature required to complete a reproductive stage [2], while ACHU accounts for the cessation of maize growth when nighttime temperatures fall below 4.4 °C and daytime temperatures drop below 10 °C [41], providing a more accurate representation of temperature impacts on phenological development. We calculated both metrics for the period from May 1 to October 31

$$CGDD = \sum_{n=i}^{N} \left(\frac{Temp_{max} + Temp_{min}}{2} - Temp_{base} \right) \tag{1}$$

$$Temp_{max} = \begin{cases} Temp_{max} \left(Temp_{opt} > Temp_{max} > Temp_{base} \right) \\ Temp_{opt} \left(Temp_{max} \geq Temp_{opt} \right) \\ Temp_{base} \left(Temp_{max} \leq Temp_{base} \right) \end{cases} \tag{2}$$

$$Temp_{min} = \begin{cases} Temp_{min} \left(Temp_{opt} > Temp_{min} > Temp_{base} \right) \\ Temp_{opt} \left(Temp_{min} \geq Temp_{opt} \right) \\ Temp_{base} \left(Temp_{min} \leq Temp_{base} \right) \end{cases} \tag{3}$$

where N is the day of year at the end of the maize growth season; n denotes the beginning of the growing season; and i represents the day of year at the start of the growing season. We set $Temp_{opt}$ and $Temp_{base}$ to 32 °C and 10 °C, respectively. $Temp_{opt}$ corresponds to the established high-temperature threshold for maize growth season, which is commonly used in $CGDD$ calculations [41].

We calculated the $ACHU$ for maize as follows [41]:

$$ACHU = \left(1.8 \times \left(Temp_{min} - 4.4\right) + 3.3 \times \left(Temp_{max} - 10\right) - 0.084 \times \left(Temp_{max} - 10\right)^2\right) \Big/ 2 \tag{4}$$

Moreover, this study calculated the difference in CGDD and ACHU between two adjacent phenological stages to quantify the temperature and cumulative heat required for the subsequent phenological stage. Their expressions are as follows:

$$CGDD_DIFF_j = CGDD_j - CGDD_{j-1} \tag{5}$$

$$ACHU_DIFF_j = ACHU_j - ACHU_{j-1} \tag{6}$$

where CGDD_DIFF and ACHU_DIFF represent the differences in CGDD and ACHU between two adjacent phenological stages. j represents the date at six phenological stages. $j-1$ represents the date at the previous stage of the jth phenological stage. Additionally, this study calculated the differences of ShortWave (ShortWave_DIFF), Pre (Pre_DIFF), and Soil

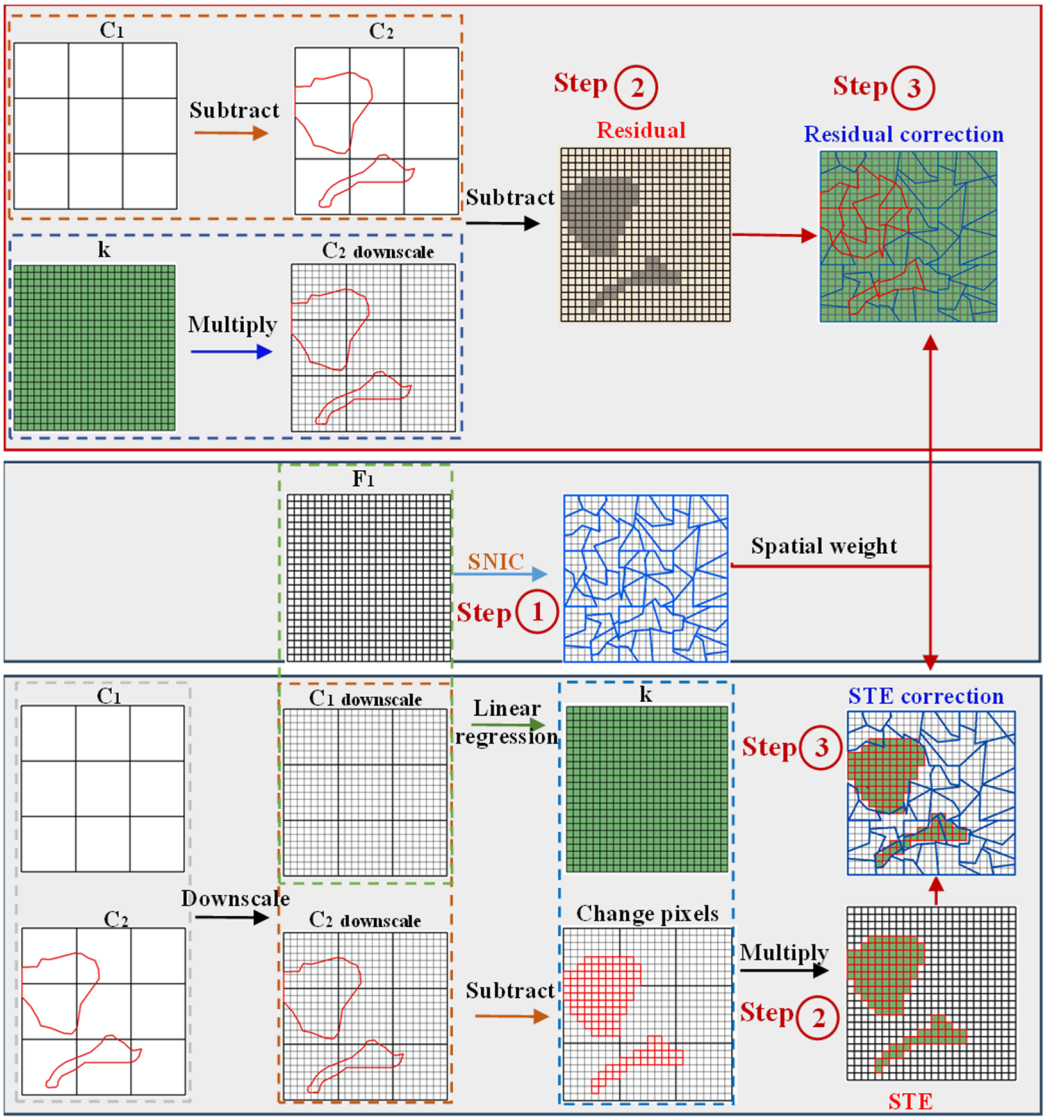


Fig. 4. Flowchart for STE and residual correction.

water (Soil water_DIFF) between two adjacent phenological stages. We further calculate these indicators as the mean values for each phenological stage to provide prior knowledge for predicting unknown phenological information.

4) Maize Distribution Data: This study utilized maize distribution maps with 30-m spatial resolution from 2001 to 2020 to mask planting areas. These maps were generated using a high spatiotemporal resolution fused dataset and time-weighted dynamic time warping. Validation with 54 281 field survey samples showed an average user's accuracy of 77.32%, a producer's accuracy of 80.98%, and an overall accuracy of 80.06% [43].

III. Methodology

This study developed a stepwise strategy to monitor six maize phenological stages from 2001 to 2020 using the GEE platform (Fig. 2). First, ground phenology observation data were aligned with maize fields using maize distribution maps, NDVI time series, and Google Earth images. Second, Blue, Green, Red, NIR, SWIR1, and SWIR2 bands were fused from Landsat and MODIS data over the 20 years using quick STF with coarse- and fine-resolution STEs and pixel-based synthesis base image pair (STEPSBI). Third, we proposed a stepwise strategy to build phenology models for vegetative growth phases (VGPs) and reproductive growth phases (RGPs) using RF regression. The derived results were validated, and uncertainties were assessed using ground phenology observations and comparisons with various phenological products.

A. Generate Phenological Detection Samples

This study generated phenological detection samples by aligning the ground phenology observation data with maize fields. However, misclassification of maize spatial distribution data could introduce noise. Thus, we developed an automated interactive method to determine the spatial location of maize fields based on their unique temporal curves during the growing season. As shown in Fig. 3, we improved the accuracy of maize field identification in three key aspects: first, the dryland

crops are maize and soybeans in the study area, soybean emergence occurs later than maize, and soybean matures earlier than maize. Therefore, when maize fields were still covered with green vegetation, the soybean fields had already turned into bare soil [Fig. 3(h) and (i)]. Second, maize grows faster than soybean at the JD, leading to a rapid rise in NDVI [red cross lines in Fig. 3(e) and (j)]. Third, after pollination, the male and female ears dry out and yellow, causing a slight decrease in NDVI [red rectangle in Fig. 3(e) and (j)]. Through the above characteristics, we guaranteed the accuracy of the maize field. Finally, we matched the fields within a 20 km radius, as previous studies indicate that comparisons in the 5–20-km range are minimally affected by spatial scales [2], which is also supported by Liu et al. [44].

B. Fusion Landsat–MODIS Data Through STEPSBI

This study employed STEPSBI [19] to fuse Landsat and MODIS images at eight-day intervals with 30-m spatial resolution from 2001 to 2020. This approach effectively recovers spatial details in heterogeneous areas during the growing season. STEPSBI utilized the optimal pixel-based image synthesis based on all available Landsat and MODIS images in each stage (Table S1), mitigating the issue of missing base images. Then, we quantified the STE of MODIS downscaling to Landsat in temporal variation. A residual term was introduced to reduce prediction errors associated with temporal variation. Finally, we corrected spatial scale and temporal errors using super-pixel segmentation results as spatial weights (Fig. 4). The specific steps are as follows:

$$\left(\tilde{F}_b, \tilde{C}_b\right) = f(S_b) \tag{7}$$

$$\left(\tilde{F}_a, \tilde{C}_a\right) = f(S_a) \tag{8}$$

where S_b and S_a are the image collections before and after the predicted time (T_2). f represents the pixel-level image synthesis operation according to "similarity" and "consistency" criteria. F and C represent the fine- and coarse-resolution image time series, respectively. a and b represent the images after and before T_2, respectively.

Then, we derived the calculation formula of STE. The preliminary blurring relationship between fine- and coarse-resolution images is modeled as follows:

$$k = F_1 / C_{1,down} \tag{9}$$

where k is the conversion coefficient between F_1 and $C_{1,down}$, which expresses the scale transformation relationship. F_1 and C_1 represent the optimally synthesized base images, while $C_{1,down}$ is obtained by cubic convolution interpolation of C_1 at reference time (T_1). If the land surface remains unchanged in space, $C_1 = C_2$, and $F_1 = F_2$, the k at T_1 and T_2 are consistent, there is no error in k, $STE = 0$, and k and C_2 could be directly predicted by F_2. However, because the same image does not exist at different times, the land cover changes from C_2 to C_1 would amplify the error of k at heterogeneous surfaces. Therefore, the STE was derived as follows:

$$STE = k * (C_{2_{down}} - C_{1,down}) \tag{10}$$

where $C_{2,down}$ is the C_2 by cubic convolution interpolation at T_2.

Next, we derived the calculation formula for the Residuals (Res), which represent the difference between coarse- and fine-resolution temporal changes, capturing the temporal error between the actual and predicted values at fine resolution in heterogeneous surfaces. Assuming that k inherits the transformation relationship between coarse- and fine-resolution data at T_2, the initially predicted image ($\tilde{F}_2$) can be expressed as

$$\tilde{F}_2 = k * C_{2,down}. \tag{11}$$

The Res term is introduced to reduce the prediction error between $\tilde{F}_2$ and predicted image (F_2)

$$Res = (C_2 - C_1) - \frac{1}{l}\sum_{i=1}^{l}(\tilde{F}_{2,i} - F_{1,i}) \tag{12}$$

where C_2 and C_1 are the coarse-resolution images at T_2 and T_1, respectively. l is the fine-resolution pixel number contained in a coarse-resolution pixel. $\tilde{F}_{2,i}$ and $\tilde{F}_{1,i}$ are the fine-resolution images of the ith pixel at T_2 and T_1, respectively.

Furthermore, we applied STE and Res correction, which are crucial for improving the fusion accuracy over heterogeneous surfaces. The specific steps are shown in Fig. 4, and their spatial weights are distributed equally according to the segmentation results of simple noniterative clustering, under the assumption that the temporal change trend of ground objects within a super-pixel is uniform, thus reducing the uncertainty caused by spatially heterogeneous surfaces. The prediction of Landsat images could be expressed as follows:

$$F_{2,Oi} = F_1 + \frac{1}{p}\sum_{j=1}^{p}\left[STE_{down,Oij} + Res_{down,Oij}\right] \tag{13}$$

where Oi is the ith super-pixel object. p represents the fine pixel number in the super-pixel object, and Oij refers to the jth pixel in ith super-pixel. $STE_{down,Oij}$ is the jth pixel in the ith super-pixel of the downscaled STE image. And $Res_{down,Oij}$ is the jth pixel in the ith super-pixel of the downscaled Res image. For a detailed description, refer to [19]. Moreover, we synthesized the data for each phenological stage using the median of the fused remote sensing images. We randomly selected 1000 pixels in the blue, green, red, NIR, SWIR 1, and SWIR 2 at six phenological stages from 2001 to 2020. We compared their reflectance values with the corresponding Landsat observations for validation. Finally, the fused Landsat images and the raw Landsat images were then median-synthesized according to the duration of each phenological stage to ensure comprehensive spatial coverage for every phenological stage in each year.

C. Selection of Input Variables by Frequency Distribution

This study calculated the Gaussian probability density distribution to assess the separability across various phenological stages (Fig. 5). Variables with two or more peaks, such as ACHU and CGDD, exhibited six distinct probability distribution peaks, indicating better separability than band reflectance and spectral indices. We standardized each variable to eliminate dimensionality effects for better comparison. For V3

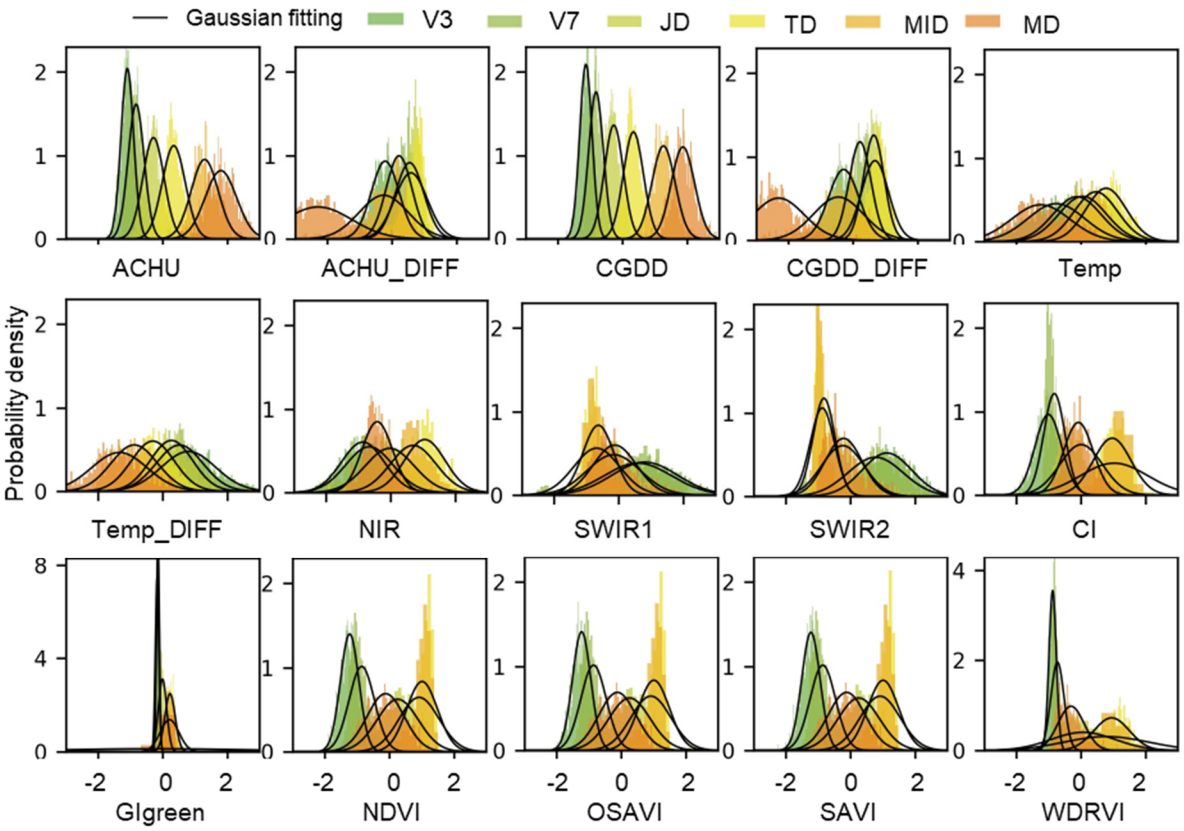


Fig. 5. Probability density distribution of variables in different phenological stages.

and V7, the densities of ACHU, CGDD, NIR, and spectral indices ranged from −2 to 0, while Temp_DIFF, SWIR1, and SWIR2 varied from 0 to 2, demonstrating the complementarity of these variables in reducing confusion. For JD and TD, ACHU, CGDD, and Temp_DIFF generally fell between −1 and 1, whereas band reflectance and spectral indices spanned −1 to 2. The densities of ACHU_DIFF, CGDD_DIFF, and Temp ranged from 0 to 2, highlighting their role in identifying JD and TD. Similarly, for MID and MD, the densities of ACHU, CGDD, and CI were primarily between 0 and 2, while band reflectance and spectral indices extended approximately from −2 to 2. The densities of ACHU_DIFF, CGDD_DIFF, Temp_DIFF, and Temp ranged from −3 to 1. Based on these findings, we selected these variables as inputs for model training (Fig. 5).

D. Detection Phenology by Stepwise Strategy

This study developed a stepwise strategy to improve the accuracy of multistage phenological detection. Specifically, the phenological stage is divided into two groups, including the vegetative growth stage, $S_{\mathrm{VGP}} = \{\mathrm{V3}, \mathrm{V7}, \mathrm{JD}\}$ and reproductive growth stage, $S_{\mathrm{RGP}} = \{\mathrm{TD}, \mathrm{MID}, \mathrm{MD}\}$. We modeled VGP and RGP separately for two reasons. First, spectral reflectance and vegetation index generally increase during the vegetative growth stage (VGP) while decreasing during the reproductive growth stage (RGP). These opposite trends could result in confusion of variables at various stages. Second, the importance of factors varies significantly between the two periods. VGP characterizes vegetation growth using incremental heat and temperature changes (CGDD_DIFF, ACHU_DIFF, and Temp_DIFF), whereas RGP is influenced by cumulative heat and temperature. As shown in Fig. 7, VGP primarily relies on the differential temperature and CI, while RGP relies on the cumulative temperature, CI, or NIR. Therefore, we developed two detection models in VGP and RGP, respectively, and selected the optimal combination of spectral reflectance, vegetation indices, and meteorological reanalysis factors using a recursive feature elimination approach.

For each phenological stage (s), their occurrence date (d_s) were predicted. The feature sets X_t^{VGP} and X_t^{RGP} are constructed at the VGP and RGP, respectively. The mapping relationship of stage s was expressed as f_s. The X_t^{VGP} included CGDD_DIFF, ACHU_DIFF, Temp_DIFF, CI, GIgreen, NIR SAVI, WDRVI, NDVI, OSAVI, SWIR1, and SWIR2. The X_t^{RGP} were CGDD, ACHU, Temp, CI, GIgreen, NIR, SAVI, WDRVI, NDVI, OSAVI, SWIR1, and SWIR2. In S_{VGP} and S_{RGP}, we established the RF regression models f_s^{VGP} or f_s^{RGP}. The specific expression is as follows:

$$d_s^{VGP} = f_s^{VGP}(X_t^{VGP}) s \in S_{\mathrm{VGP}} \tag{14}$$

$$d_s^{RGP} = f_s^{RGP}(X_t^{RGP}) s \in S_{\mathrm{RGP}} \tag{15}$$

where d_s^{VGP} and d_s^{RGP} are the phenological occurrence dates at VGP and RGP, respectively.

Additionally, the RF model's optimal parameters were established after numerous experiments: default values were

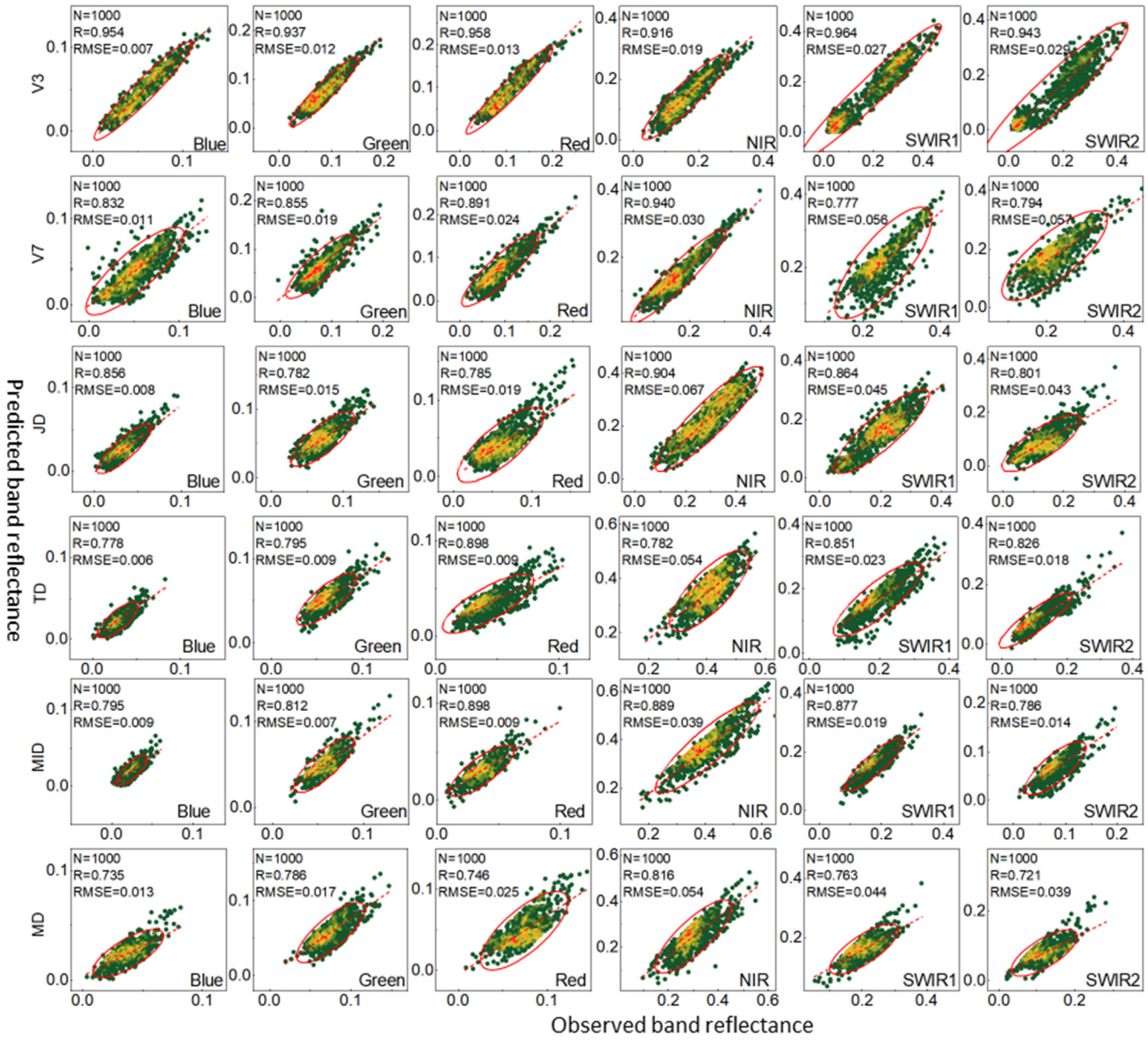


Fig. 6. Performance of the STEPSBI STF method in band reflectance at each phenological stage during 2001–2020. The red ellipses indicate the 95% confidence interval.

used for mtry (the square root of the number of variables) and unlimited node size per tree, with ntree set to 500.

E. Validation and Uncertainty Evaluation

This study evaluated the detected model and results from three aspects, including comparison with different methods and datasets, as well as the analysis of the uncertainty of the detection results.

1) Model Comparison: To assess the advantage of the proposed approach in multistage phenology detection, we selected three recent and representative baseline methods: the improved shape model fitting (SMF-S), a 1-D U-Net model, and the phenology Seq2Seq network (PSeqNet) [44], [45], [46]. They represent mainstream approaches in current multistage phenology detection. SMF-S [44] introduces local adaptive fitting for each stage. However, it relies on predefined phenological references and linear transformations, resulting in phenological associations that are overly dependent on these references, making it challenging to adapt to dynamic phenological changes under climate variability. In contrast, the 1-D U-Net [45] is a recent data-driven methods, which learn complex temporal patterns from dense optical and synthetic aperture radar (SAR) time series through convolutional and recurrent layers, providing stronger capability in modeling dynamic transitions. Nevertheless, its performance remains sensitive to training data and may encounter challenges in large-region or long-term spans. PSeqNet is a novel crop phenological monitoring method that adopts a two-stream input encoder and an autoregressive phenological decoder structure, fully considering the depen-

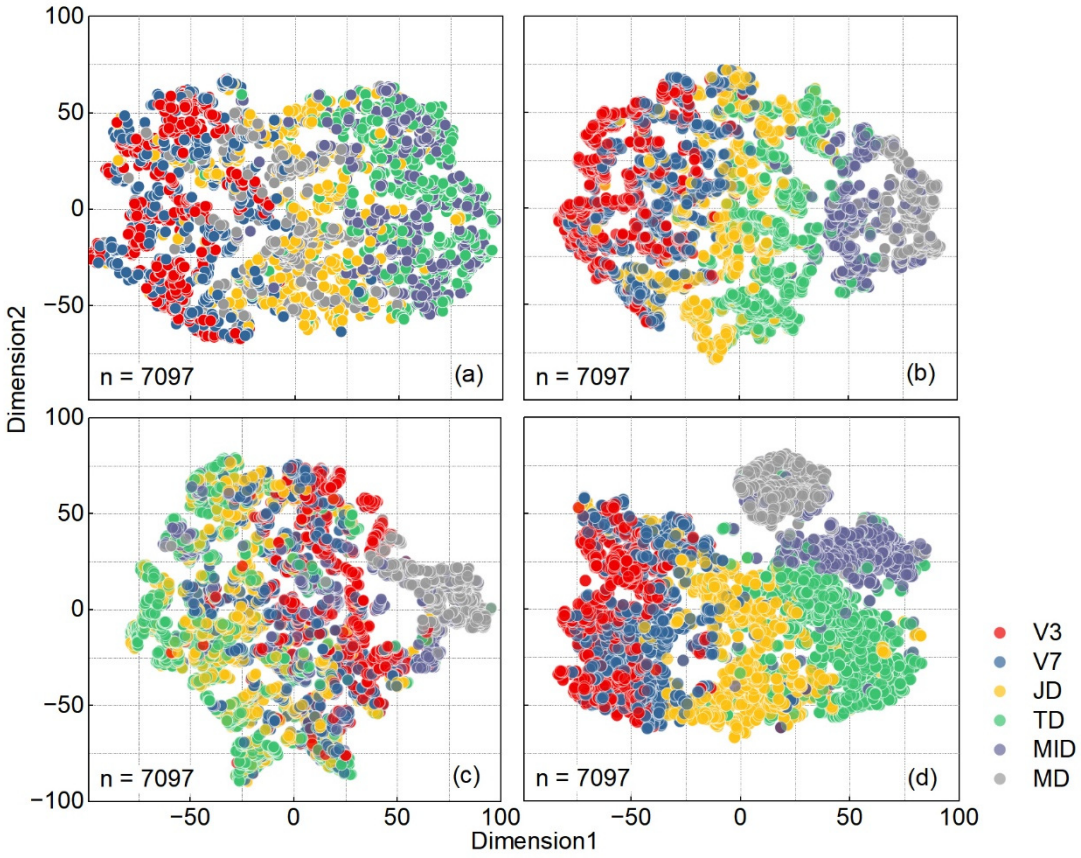


Fig. 7. Illustration of influence interaction among input variables for phenological detection: (a) 2-D space derived from spectral reflectance and vegetation index; (b) ACHU, Temp, and CGDD; (c) ACHU_DIFF, Temp_DIFF, and CGDD_DIFF; and (d) extracted from all variables in (a)–(c).

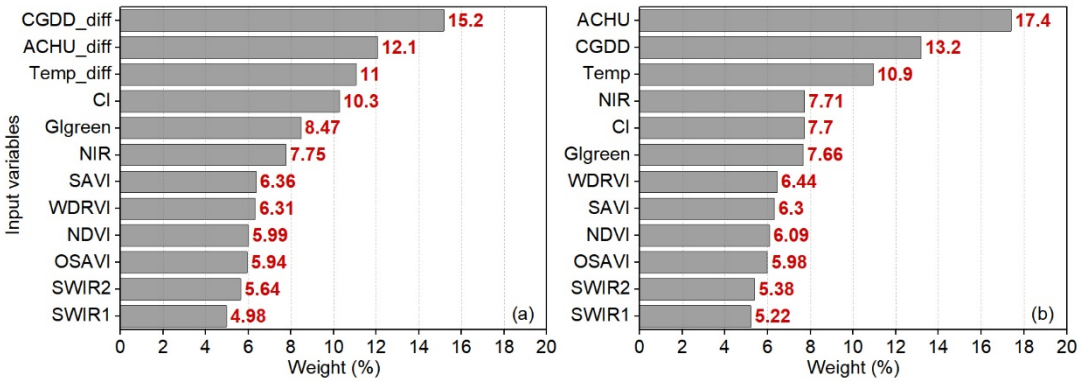


Fig. 8. Weights of different input variables. (a) and (b) Importance ranking of input variables for detecting V3, V7, and JD in the vegetative growth stage (VGP) or TD, MID, and MD in the reproductive growth stage (RGP) by stepwise strategy.

dence relationship between adjacent phenological stages [46]. Overall, shape model fitting and deep learning methods provide a fairer and more comprehensive performance comparison. They highlight the novelty and effectiveness of our proposed approach in modeling complex multistage and long-time series phenology under climate variability.

As shown in Fig. 1, we set the model training and validation ratio at 3:1 for each year from 2001 to 2020. The validation site did not participate in training and served as an independent validation. Specifically, each phenology stage was averaged within a 20 × 20 km grid and compared to ground observations [12], [44]. In the SMF-S, the reference curves used in this study were averaged from the model training sites to ensure a fair comparison (Fig. 1). The sites for model validation were evaluated, and the model training stations in the study area were considered suitable in this case (Table S2). Consistent with most studies, the SMF-S method is applied based on the vegetation index that reflected canopy greenness [52], specifically WDRVI in this case. For the 1-D U-Net, the training stations and validation stations are consistent with our proposed approach. Moreover, the accuracy of detection was assessed using root-mean-square error (RMSE) and the

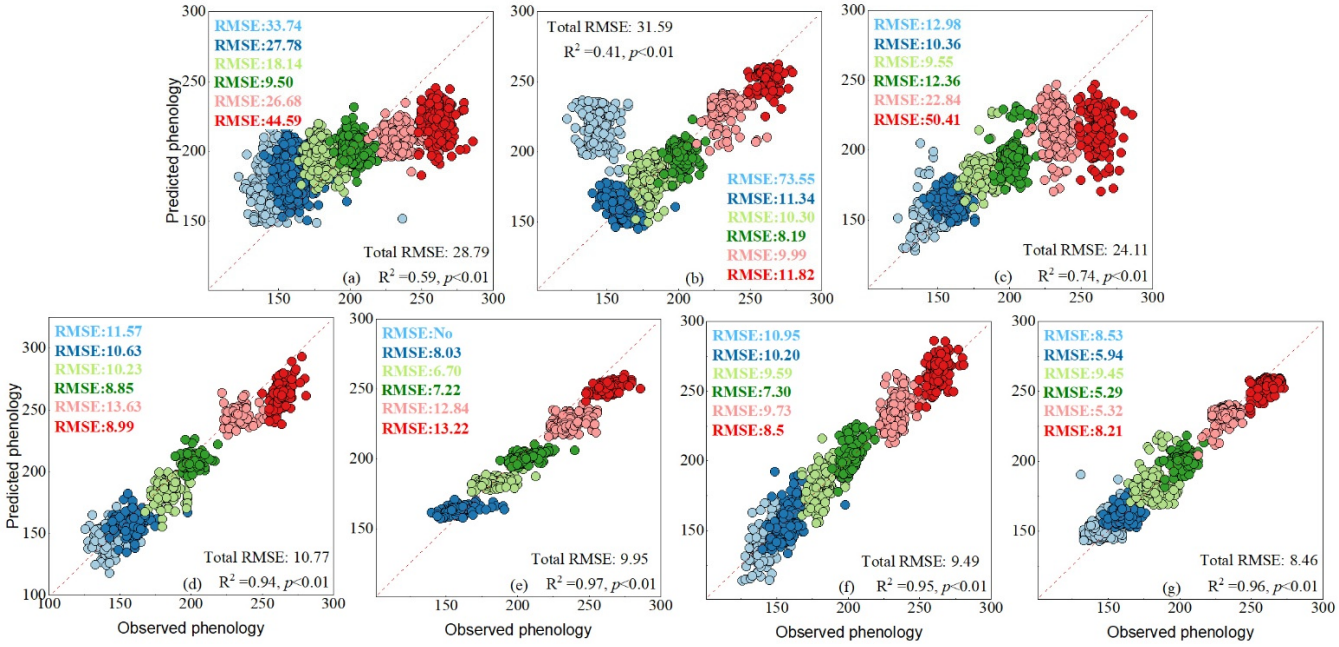


Fig. 9. Comparison of accuracy in various phenology detection approaches: (a) relationship between ground observations and detected phenology by spectral reflectance and vegetation index as inputs; (b) extracted from cumulative heat units (ACHU), temperature (Temp), and CGDDs as inputs; (c) extracted from the difference of heat units (ACHU_DIFF), temperature (Temp_DIFF), and cumulative growing degree day (CGDD_DIFF) between two adjacent phenological stages as inputs; and (g) extracted from all variables in (a)–(c) as inputs by stepwise strategy. (d) 1-D U-Net. (e) PSeqNet. (f) SMF-S. The p represents the significance level.

coefficient of determination (R^2). The total RMSE (Fig. 9) was calculated from predicted and recorded dates in all phenological stages, providing an overall assessment.

2) Dataset Comparison: We compared the generated datasets with existing products generated by conventional approaches. Specifically, GLASS 1 km [27] combined both the inflection- and threshold-based methods to identify the green-up, heading, and MD of major crops. The Landsat 30 m [12] utilized a dynamic threshold method, defining V3 as the date corresponding to 10% of the EVI amplitude during the green-up phase, and MD as 50% of the EVI amplitude during the green-down phase. As well as, MODIS 500 m [47] utilized the threshold method to detect vegetation phenology. Then, we compared their spatial details (Fig. 9) and temporal trends (Fig. 10).

3) Uncertainty Analysis: To assess the uncertainty of phenological estimates, the ground-based phenological observation range was defined as the interval between the earliest and latest recorded dates for each stage, utilizing all available ground-based phenological records from 2001 to 2020 in Northeast China. For each stage (e.g., V3, V7, JD, TD, MID, or MD), the minimum and maximum dates across all stations and years were used to determine the observation range. Estimates falling outside the interval were considered incorrect, while those within the range were regarded as correct. The ratio of correct to incorrect detections is then calculated on a yearly basis. Notably, this approach does not require a one-to-one correspondence between pixels and observation stations, as the range provides a regional reference to identify estimates significantly earlier or later. This strategy effectively filters out extreme deviations while mitigating potential bias arising from the sparse and uneven spatial distribution of ground stations.

IV. Results

A. Performance of the STEPSBI STF Method

The STEPSBI STF approach demonstrated robust performance across different spectral bands in Northeast China. Fig. 6 represents the overall fusion performance from 2001 to 2020. Across six phenological stages, the fused six-band reflectance achieved the highest mean R during the V3, followed by V7 (0.85), MID (0.84), JD (0.83), TD (0.82), and MD (0.76) (Fig. 6). The approach successfully caught signals from green vegetation and soil background and performed well in detecting signals associated with crop yellowing. Moreover, the mean R of the band reflectance across the six phenological stages ranged from 0.81 to 0.88 (Table SI), indicating excellent fusion accuracy.

B. Improved Sample Representativeness by Interaction Influence of Inputs

The stepwise strategy significantly enhanced sample representativeness by utilizing optimal variable combinations for VGP and RGP (Fig. 7). We visualized phenological stage distributions using the t-distributed stochastic neighbor embedding (t-SNE) approach [48], which transformed high-dimensional input variables into two dimensions. As shown in Fig. 7(a), the clusters for V3, V7, JD, TD, MID, and MD overlapped considerably in the feature space of spectral reflectance and vegetation indices. In contrast, phenological stages were

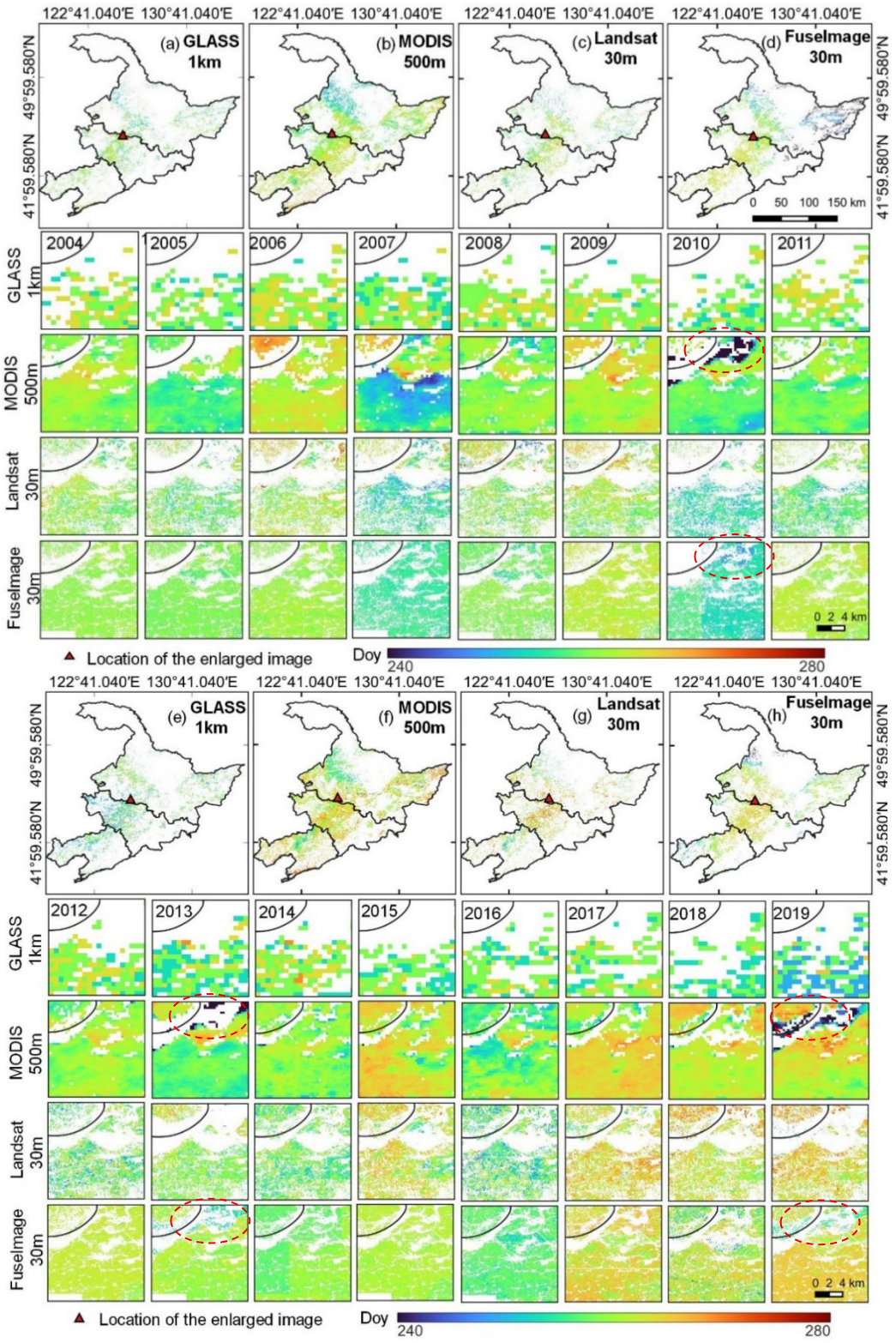


Fig. 10. Comparison of the MD was derived from the GLASS, the MODIS (mid-green-down date), the Landsat, and the fused image from 2004 to 2019. (a)–(d) MDs of the four datasets in 2011. (e)–(h) MDs of the four datasets in 2019. The enlarged area is the location of the red triangle.

more clearly separated in the feature spaces of ACHU_DIFF, Temp_DIFF, and CGDD_DIFF, as well as ACHU, Temp, and CGDD [Fig. 7(b) and (c)]. This suggests that temperature and heat accumulation derived from meteorological data were more effective in distinguishing phenological stages than vegetation indices and band reflectance. Consequently, this study selected the optimal variable combinations for VGP and RGP, integrating spectral reflectance, vegetation indices, and temperature-derived factors [Fig. 7(d)]. The results indicated the interaction influence of inputs beneficial to improve sample

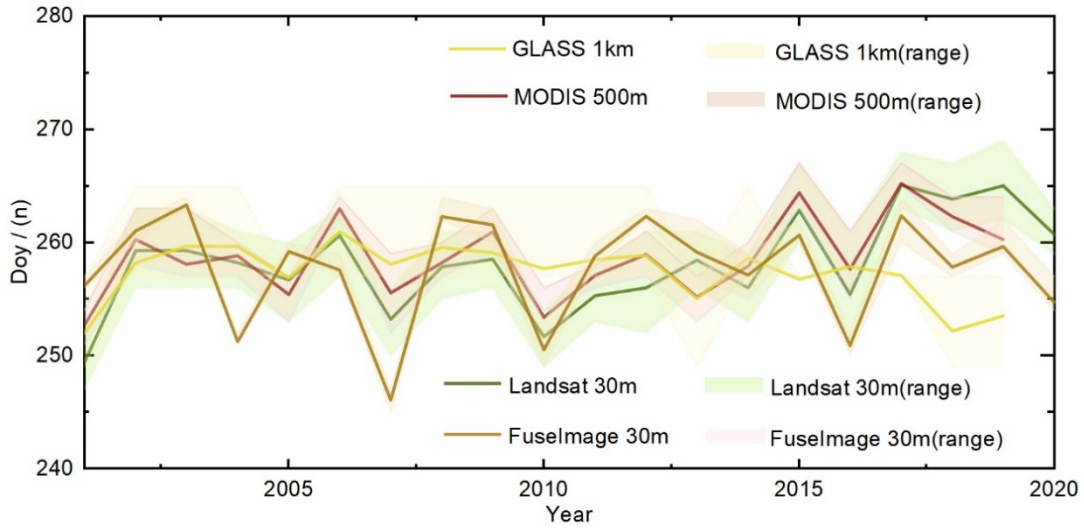


Fig. 11. Temporal trend of the MD was derived from the GLASS, the MODIS (mid-green-down date), the Landsat, and the fused image from 2001 to 2020 at the regional level. Solid lines are the mean MDs at the regional level, and shadowed frames indicate the range of MD within the 25th and 75th quantile levels.

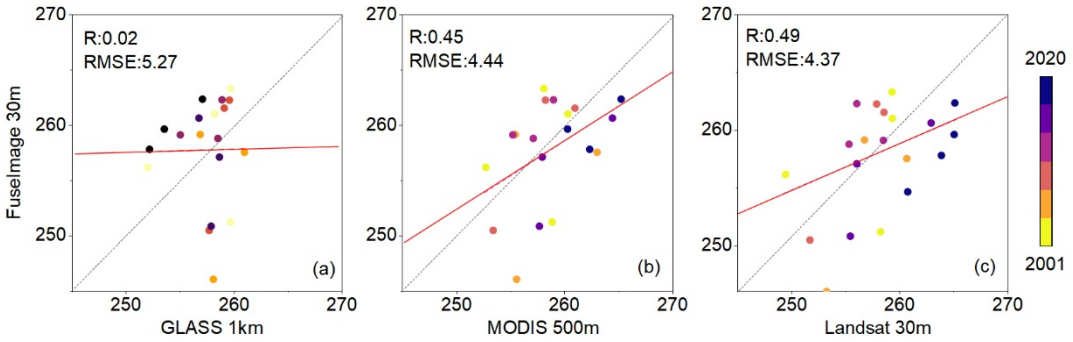


Fig. 12. Cross-validation of phenology extracted from fused images with that of GLASS 1 km (a), MODIS 500 m (b), and Landsat 30 m (c).

representativeness and facilitate the identification of multistage phenology.

Furthermore, this study assessed the importance of input variables [49]. The weight values of each input variable were standardized to sum to 1. The results revealed that the top five variables with the highest weights in VGP were CGDD_DIFF, ACHU_DIFF, Temp_DIFF, CI, and GIgreen, whereas in RGP were ACHU, CGDD, Temp, NIR, and CI (Fig. 8). The top three variables consistently included CGDD_DIFF, ACHU_DIFF, and Temp_DIFF with the weights ranging from 10.9% to 17.4%. Among the reflectance bands, NIR exhibited higher weights, with values of 7.75% and 7.71% in VGP and RGP, respectively. CI and GIgreen displayed higher weights among the vegetation index, ranging from 7.66% to 10.3%. Notably, CI outperformed other vegetation indices due to its capacity to capture soil–canopy interactions throughout maize growth. For example, during the V3 stage, when leaves are narrow, most of the soil remains exposed; during TD and MID, the canopy closes and completely covers the ground; and at MD, senescent leaves coexist with soil signals. Unlike vegetation indices, which saturate under dense canopies, CI is sensitive to the temporal variability in the soil background [39]. This property demonstrates its superior capacity to distinguish various phenological stages.

C. Comparison With the Ground Observation Data

The stepwise strategy achieved a total RMSE of 8.46 days, which integrated spectral reflectance, vegetation indices, and temperature-derived factors. It significantly outperformed RF models that only used spectral reflectance (RMSE: 31.59 days) or temperature-derived factors (RMSEs: 24.11 and 28.79 days) in detecting spring maize phenology (Fig. 9). When ACHU, Temp, and CGDD were used as inputs [Fig. 9(b)], RMSEs from TD to MD ranged from 8.19 to 11.82 days, which were smaller than those from V3 to JD (10.30–73.55 days). Conversely, using ACHU_DIFF, Temp_DIFF, and CGDD_DIFF as inputs [Fig. 9(c)] produced RMSEs from V3 to JD ranging from 9.55 to 12.98 days, outperforming those from TD to MD (12.36–50.41 days). For the RF model based solely on spectral reflectance and vegetation indices [Fig. 9(a)], TD showed the highest concordance with ground observations. However, significant overestimations occurred from V3 to JD, accompanied by clear underestimations at MID and MD. This confusion could be attributed to the rise and decline of spectral signals during growth, resulting in similar values

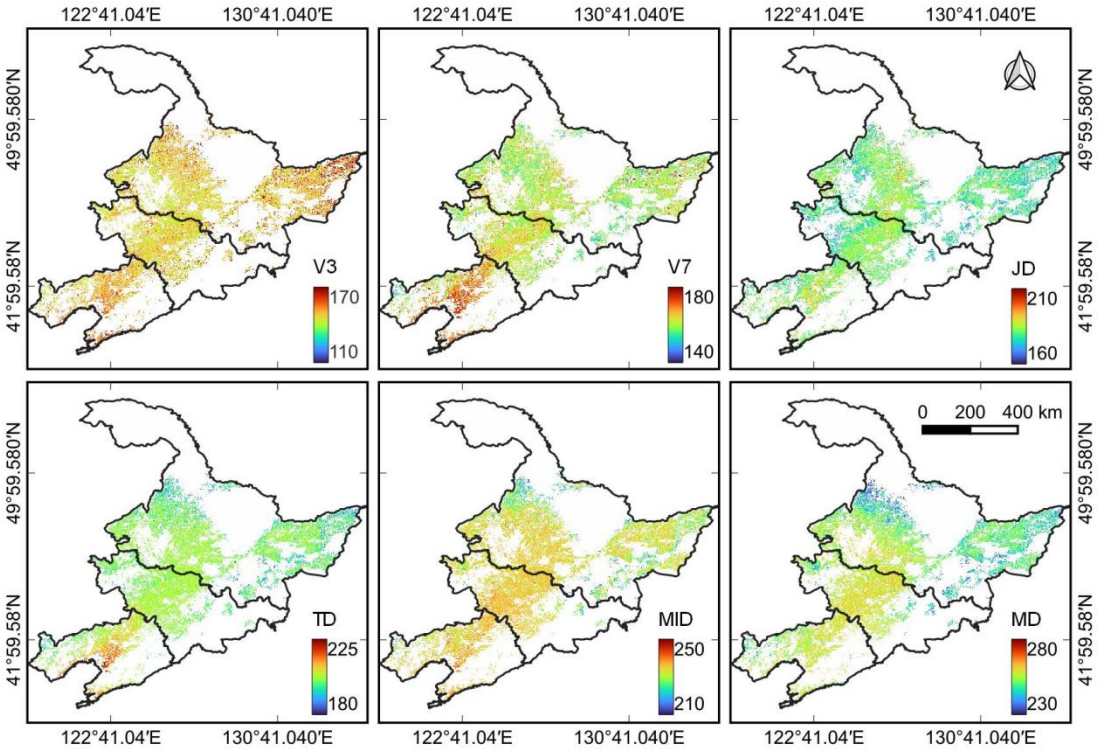


Fig. 13. Spatial patterns of annual averages of six phenological dates from 2001 to 2020.

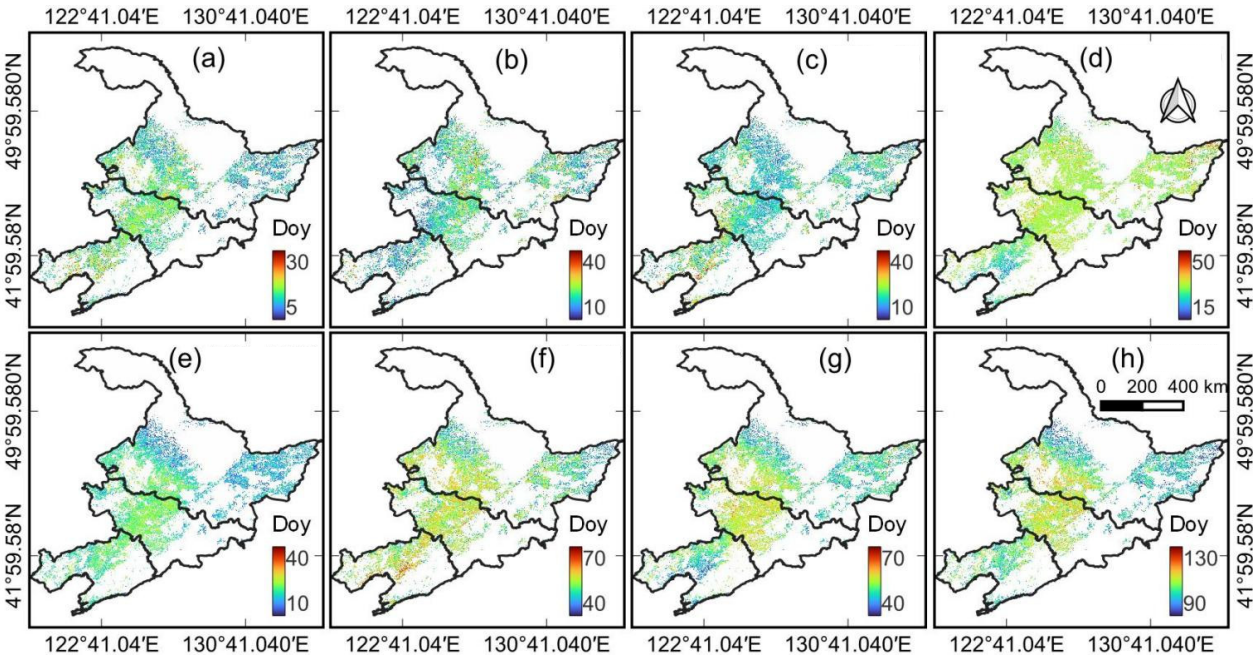


Fig. 14. Spatial distribution of time intervals between adjacent phenological stages. The time intervals are shown from (a) V7–V3, (b) JD–V7, (c) TD–JD, (d) MID–TD, and (e) MD–MID. (f) VGP refers to the vegetative growth period from TD to V3. (g) RGP denotes the reproductive growth period from MD to TD. (h) WGP represents the whole growth period from MD to V3.

across different phenological stages. To address this, this study proposed a stepwise strategy by constructing separate VGP and RGP models, achieving optimal accuracy across six phenological stages with an R^2 of 0.96 and an RMSE of 5.29 days [Fig. 9(f)].

Furthermore, we assessed three contemporary approaches for phenology detection. As shown in Fig. 9, the stepwise strategy has superior monitoring accuracy, achieving a total RMSE, that is, 2.31, 1.49, and 1.03 days lower than those of the 1-D U-Net, PSeqNet, and SMF-S, respectively. This improvement could be attributed to the physiological characteristics inherent in maize growth as well as regional-scale long-term monitoring. Subtle spectral differences, heterogeneous environments, and interannual variability complicate

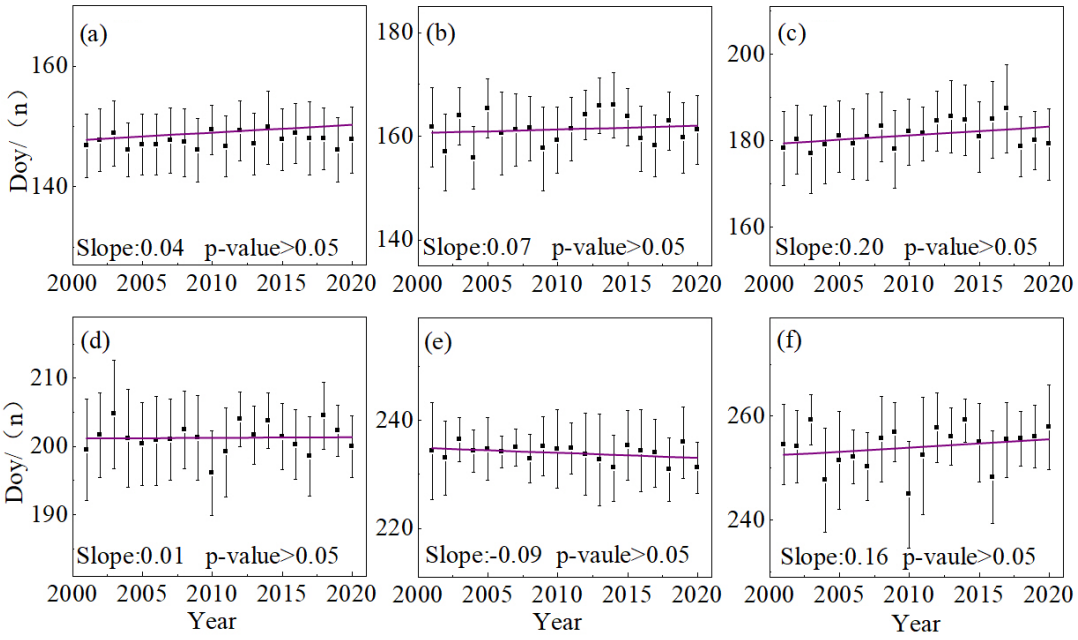


Fig. 15. Temporal trends of six phenological stages from 2001 to 2020. The data were independently fit for two segments (i.e., 2001–2010 and 2011–2020) due to distinct temporal trends in each period. Annual variability for the study area is defined as $(x - \mu)/\sigma$, where x represents the value of each phenological stage in a year, and μ and σ are the mean and standard deviation of each phenological stage, respectively. (a) V3. (b) V7. (c) JD. (d) TD. (e) MID. (f) MD.

stage separation. Our approach effectively delineates vegetative and reproductive phases while integrating both differential and cumulative temperature indices alongside spectral features. This enables it to capture short-term transitions as well as long-term accumulation effects.

Besides, maize phenology progresses sequentially, exhibiting strong physiological dependencies across various stages. PSeqNet adopts a sequential monitoring framework, in which the estimated phenological date of a previous stage is used as an input for the subsequent stage. While this design effectively captures phenological continuity, it also makes the model susceptible to error accumulation along the phenological sequence, resulting in larger RMSE at MID (12.84 days) and MD (13.22 days). In contrast, the stepwise strategy estimates each phenological stage independently using temporally aggregated features, thereby avoiding interstage error propagation. However, when using eight-day composite imagery, these transitional signals may be further smoothed, leading to relatively larger estimation errors for JD (9.45 days). Moreover, the SMF-S benefits from local adaptation enhancements, but it remains reliant on predefined phenological references, which restrict its adaptability under changing climates. The U-Net exhibits a robust capacity for modeling complex dynamics. However, its performance is sensitive to training data and proves less reliable for cross-regional or long-term applications. These findings confirm that the proposed strategy not only achieves higher accuracy but also offers enhanced adaptability for multistage and long-term phenology monitoring at a regional scale.

D. Comparison With Different Phenology Datasets

The MD derived from the fused image exhibits strong spatial consistency with Landsat, MODIS, and GLASS datasets, while providing enhanced spatial detail (Fig. 10). MD was selected for comparison because it is the only phenological stage shared across three datasets. As shown in Fig. 10, MD has significant advancements in 2010, 2013, and 2019 (red boxes), and these variations are successfully captured by fused-based datasets and MODIS 500 m. Furthermore, the MD derived from the fused dataset is smoother than that of the Landsat 30-m dataset and reveals finer spatial details compared to the GLASS 1-km and MODIS 500-m datasets.

The MD derived from the fused-based image displayed a consistent temporal trend compared to the Landsat, MODIS, and GLASS datasets (Fig. 11), showing pronounced interannual fluctuations. For instance, MD in 2007 and 2016 advanced noticeably, whereas in 2012, it was significantly delayed. Furthermore, cross-validation results indicated that the RMSEs between our results and different products are 5.24 days (GLASS 1 km), 4.44 days (MODIS 500 m), and 4.37 days (Landsat 30 m), respectively (Fig. 12). Differences in phenological dates could be attributed to detection approaches and spatial scales. For instance, the GLASS 1-km datasets define MD as the first-order derivative of the leaf area index (LAI) time-series curve, where the negative value has the largest absolute magnitude [27]. The MD derived from MODIS 500 m and Landsat 30 m corresponds to 50% of the segment EVI2 amplitude within a cycle [12]. Additionally, the greater the spatial scale difference, the larger the observed RMSE is.

E. Spatiotemporal Analysis in Sequential Progression of Phenology

1) Spatial Patterns: V7, TD, and MD exhibited greater spatial variability than V3, JD, and MID from north to south (Fig. 13). The northern and southern regions experienced the V3 stage significantly later than the central region, which

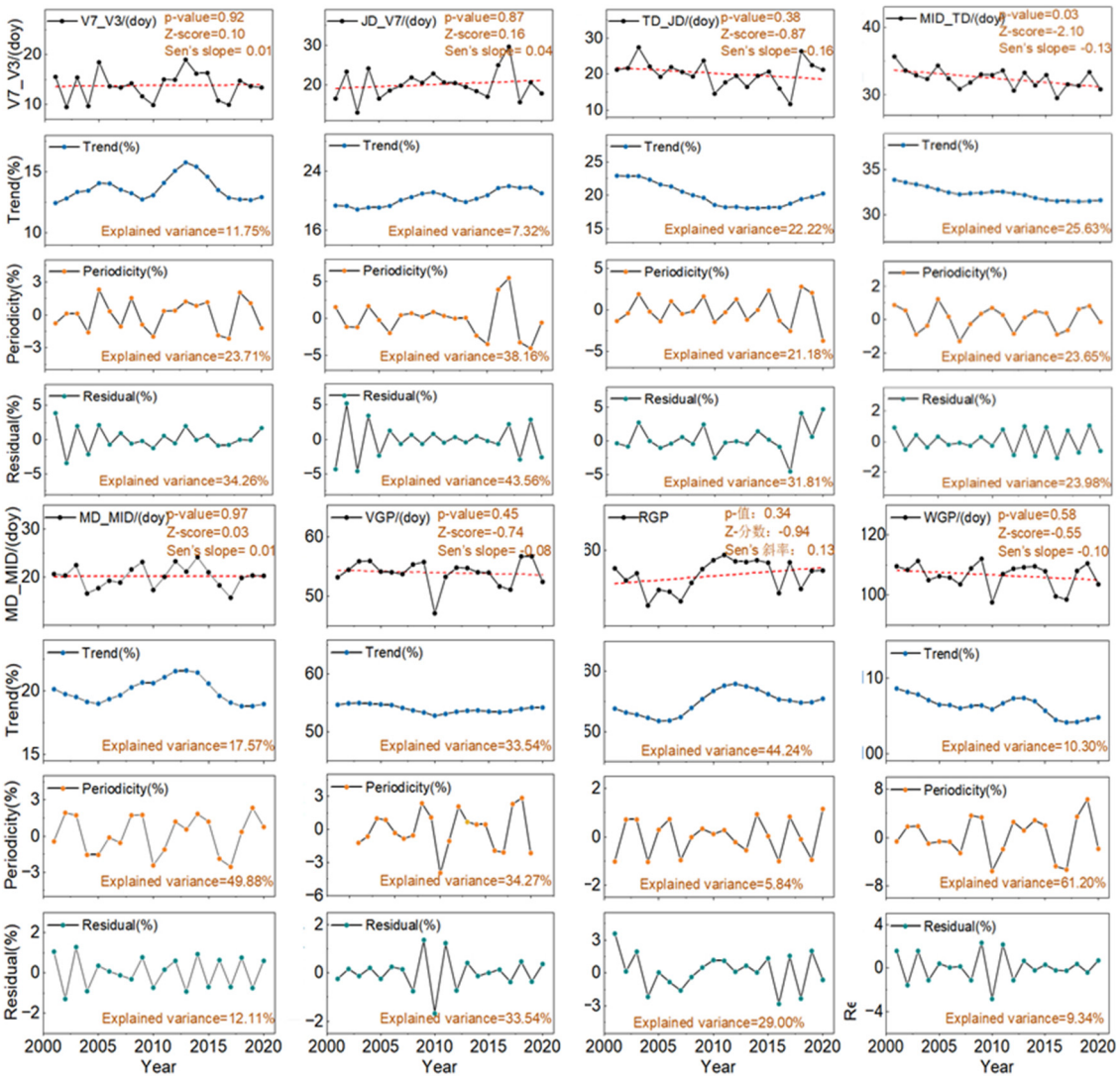


Fig. 16. Time-series trend of various growth periods. The overall trend of growth periods is described using three indicators from the Mann–Kendall test. A Z-score greater than 0 indicates an increasing trend, while a Z-score less than 0 indicates a decreasing trend. Sen's slope represents the annual rate of change in growth periods, and the p-value indicates the statistical significance of the trend. The Mann–Kendall test was performed at a 0.05 significance level. SSA decomposed the historical time series of growth periods into three components: long-term trend, periodicity, and residuals. The explained variance was calculated to quantify each component's contribution to the total variability of the data. Explained variance is the proportion of a component's variance relative to the total variance of the growth periods.

is influenced by temperature, crop variety, and management practices. For instance, early-maturing varieties are commonly used in the north to offset low temperatures and late sowing dates. Furthermore, to explain the spatial variability, this study extracted the differences between adjacent stages [Fig. 14(b)]. Although V7 and TD displayed significant spatial variations, the time for V7 and TD to transition to JD and MID increases from south to north, resulting in reduced spatial variation in JD and MID. This pattern indicated that the spatial distribution of phenological stages evolves as crops progress. Interestingly, the spatial distribution of the V3 stage closely mirrors that of the WGP. Specifically, regions in the north and south where V3 occurs later tend to exhibit shorter WGPs [Fig. 14(h)]. This spatial pattern reflects different maize cultivars adapted to local climatic conditions.

2) Temporal Trend: The start times of V3, V7, JD, TD, and MD exhibited a delayed temporal trend, whereas MID displayed an advanced trend, though not statistically significant (Fig. 15). This pattern is not completely consistent with previous research, which could be attributed to variations in spatial scale, analysis year, crop type, and management techniques [44], [50], [51]. To further analyze temporal patterns, we calculated the difference between adjacent stages and applied singular spectrum analysis (SSA) to decompose the temporal variations into trend, periodic, and residual components. The time interval from V3 to V7 exhibited statistically insignificant variance, with residual components accounting for 34.26%, indicating that this period was primarily fluctuating change. The time intervals from the V7 to JD, JD to TD, and MID to MD showed similar interannual fluctuation changes. The Sen's slopes for the intervals from V3 to V7, V7 to JD, and MID to

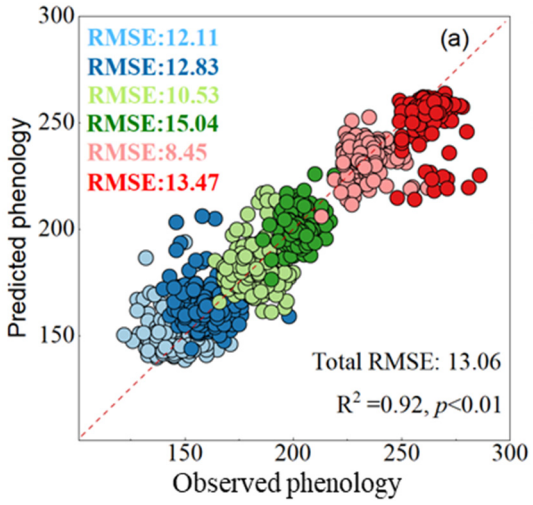


Fig. 17. Relationship between field-observed phenology and the estimated phenology by the raw images.

MD were 0.01, 0.04, and 0.01, respectively (p-value > 0.5), indicating a delayed trend. This could be linked to temperature variations in early spring, as extremely low temperatures slow the increase in land surface temperature, thereby prolonging the period required for plants to accumulate sufficient effective thermal units. During MID to MD, increased precipitation frequency in September led to a slight decrease in grain moisture content, indirectly prolonging maturity. These temporal variations partially reveal the delayed trends in V7, JD, and MD. The Sen's slope for the interval from TD to MID was −0.13 (Fig. 16), indicating that this interval from TD to MID shortened by 1.3 days per decade, resulting in an earlier MID.

Furthermore, in VGP, the periodic component accounted for the largest variance proportion at 34.27%, accompanied by trend and residual components. In RGP, the trend components exhibited the highest variance proportion at 44.24%. Notably, the WGP showed the greatest variance proportion in the periodic component (61.20%), which was primarily driven by VGP and the stages between MID and MD (Fig. 16).

V. Discussion

A. Advantages of Data Fusion to Phenology Detection

This study utilized fused images and raw images to generate the six phenological stages of spring maize from 2001 to 2020. The STEPSBI algorithm was used to generate remote sensing images for each stage. We then used both raw images and fused images for independent model training to evaluate the impact of the fused data. The results indicated that the model with fused and raw imagery achieved a lower total RMSE (8.46 days) [Fig. 9(f)] compared with the model trained using only raw imagery (13.06 days) (Fig. 17). Moreover, the RMSE for each phenological stage (5.29–9.45 days) [Fig. 9(f)] was lower than that of the raw image-based model (8.45–15.04 days) (Fig. 17). This improvement could be attributed to the fused images capturing the complex relationships among phenological stages, which were not fully learned when using only raw images for training. Furthermore, the incorporation of fusion images enhances the diversity of training samples. In summary, the fused data not only compensated for the missing raw images but also improved the model's accuracy and robustness, making it highly valuable for long-term crop phenology monitoring.

B. Importance of the Interaction Influence in Input Variables

Numerous studies have utilized time-series data, such as EVI, NDVI, and WDRVI, to detect vegetation phenology [8], [12], [52]. However, these studies often rely on subjectively established thresholds and inflection points to identify the start and end of the growing season, which can lead to significant disparities between remote sensing-derived and agronomic phenology. Although shape model matching methods have been successfully applied to detect multistage crop phenology [1], [44], [53], factors such as ground observation quality, site selection, and specific years directly impact detection accuracy. Consequently, these approaches are rarely used for historical dynamic monitoring, resulting in a gap in multistage and long-term phenological assessments.

This study proposed a stepwise strategy to monitor phenology, which integrates the complementarity among spectral reflectance, vegetation index, and temperature factors and overcomes the redundancy of inputs. The results indicated that separability among various phenological stages in the feature space of ACHU_DIFF, Temp_DIFF, and CGDD_DIFF, as well as ACHU, Temp, and CGDD, was higher than that in the feature space of spectral reflectance and vegetation indices [Fig. 7(a)–(c)]. Furthermore, integrating temperature-derived factors, spectral reflectance, and vegetation index within VGP and RGP enhanced the separability of various stages, indicating that the interaction among variables improves the representativeness of phenological samples.

Moreover, the weights of temperature-derived factors in VGP and RGP consistently rank among the top three (Fig. 8). This finding aligns with previous research confirming the contributions of GDD, Temp, and ACHU to phenology [41], [54], [55]. For instance, Ramirez-Parada et al. [54] demonstrated that using CGDD to simulate crop growth processes enabled highly accurate detection of wheat phenology. In contrast, this study revealed that GDD_DIFF, Temp_DIFF, and ACHU_DIFF were particularly effective for identifying VGP, while GDD, Temp, and ACHU played a more prominent role in detecting WGP. Interestingly, CI has the highest-weighted variables among the spectral indices, which could be a result of varying soil and vegetation proportions during different crop growth stages, leading to differences in surface texture [56]. This suggests that incorporating relative changes in soil-vegetation ratios, along with texture information, into phenology detection frameworks is beneficial to improve model accuracy.

C. Effect of the Multistage Phenology on Its Spatiotemporal Trends

This study focused on the effect of multistage phenology on spatiotemporal trends [57], in contrast to previous studies that emphasized individual phenological stages [23],

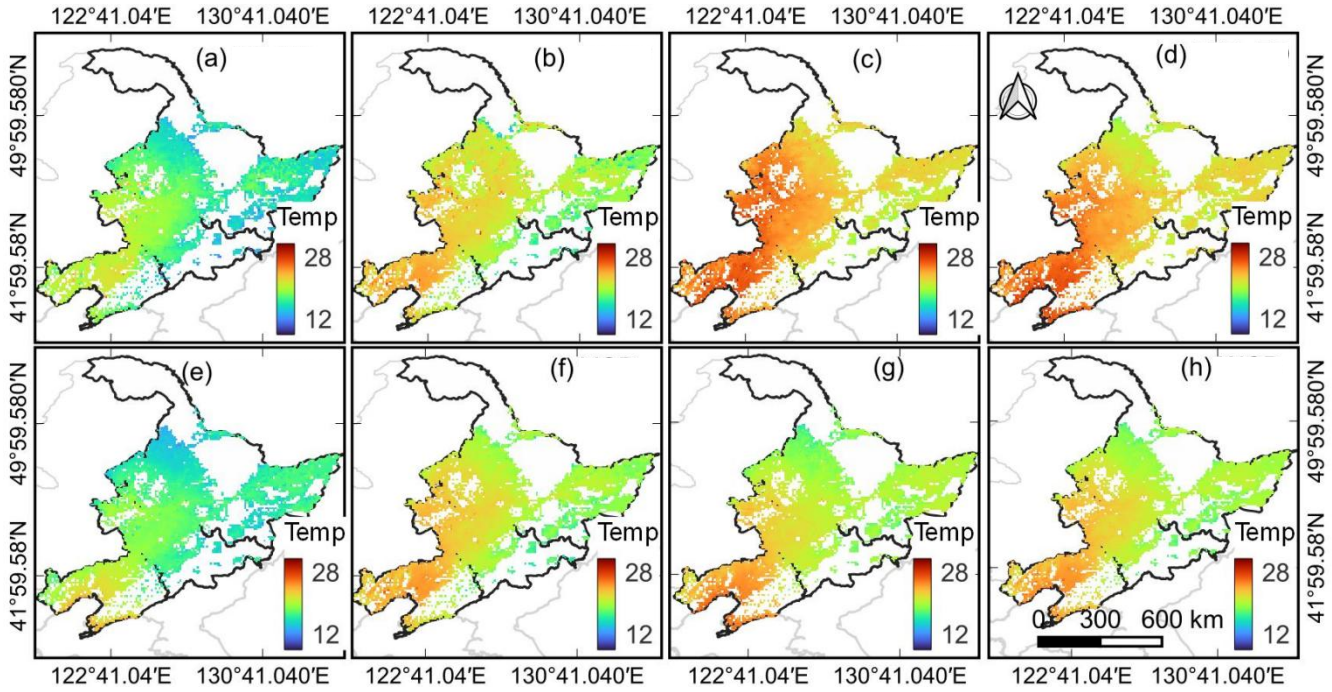


Fig. 18. Spatial patterns of annual average temperature at six phenological stages from 2001 to 2020. (a) V7–V3. (b) JD–V7. (c) TD–JD. (d) MID–TD. (e) MD–MID. (f) VGP. (g) RGP. (h) WGP.

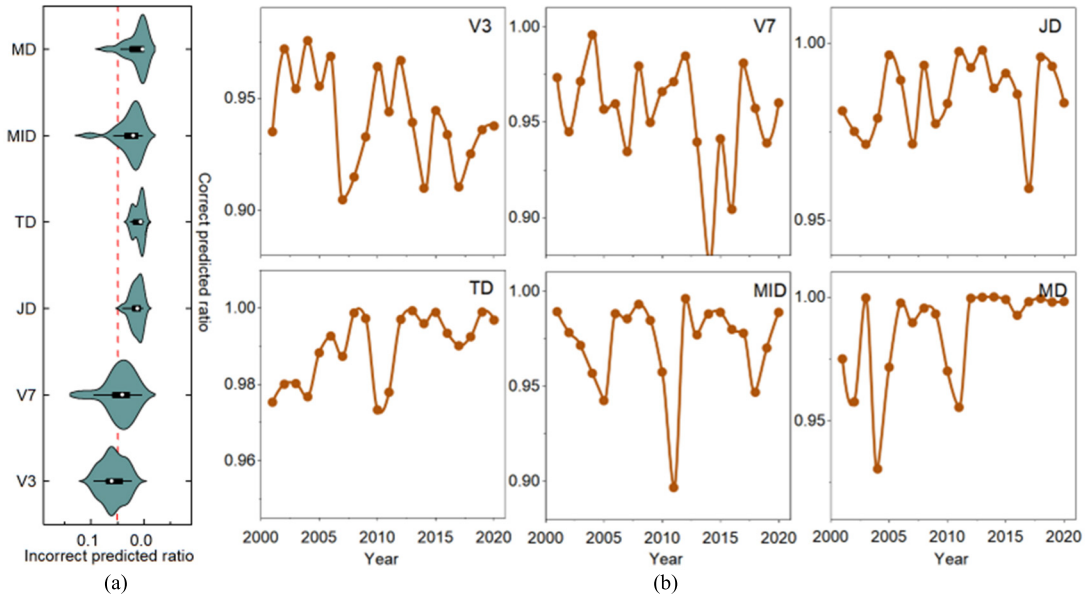


Fig. 19. Ratio of (a) incorrect to (b) correct predictions for the six key phenological stages.

[58], [59]. Multistage phenology and duration between adjacent stages reflect spatial variations among different maize cultivars, which are largely influenced by temperature thresholds that trigger subsequent stages. For instance, early-maturing and late-maturing varieties in the study area require CGDDs of approximately 2300 °C and 3000 °C, respectively [60]. To better understand the factors influencing phenological progression, we calculated the average temperature between adjacent stages. As shown in Fig. 18, the spatial distribution of temperature decreased from southwest to northeast during various phenological intervals and growth periods. Except for the interval from V3 to V7, the durations between adjacent phenological stages were inversely related to the spatial distribution of temperature in the corresponding periods (Fig. 14). Specifically, the time intervals from V7 to JD, JD to TD, TD to MID, and MID to MD showed an inverse relationship with temperature, indicating that longer temperature accumulation is required to process to the next stage. In contrast, the intervals from V3 to V7 align with temperature, which could be attributed to variations in planting schedules from north to south regions. Furthermore, although the spatial distribution of temperature was similar during VGP and the RGP, the

spatial patterns of their durations exhibited opposite trends in the southern part of the study area. Specifically, the VGP lasted longer in the Liaoning Province, while the RGP was shorter. In contrast, the Jilin Province experienced the longest durations for both VGP and RGP, whereas the northern region, Heilongjiang Province, showed relatively short durations for both phases, reflecting the spatial distribution of maize cultivars.

D. Uncertainties and Limitations

This study has several limitations in long-term phenological monitoring. First, fused images could introduce spatial errors when missing images exceed two months due to heavy cloud cover (over 60%) [19]. Second, sensor differences among ETM, TM, and OLI also affected the detected results, despite corrections applied across multiple sensors. To quantify these uncertainties, we calculated the proportion of correctly detected phenological stages in each year. The results indicated that the correct detection rate exceeded 90% over the 20 years [Fig. 19(b)], the incorrect detection proportion for V3 and V7 was approximately 5%, and other phenological stages had an error rate below 5% [Fig. 18(a)], demonstrating the reliability of the dataset. Additionally, unlike Niu et al. [12], who used a one-year maize layer to mask the multiyear datasets, this study updated the maize layer of each year according to the latest maize classification dataset, more accurately capturing the annual maize phenology information.

VI. Conclusion

This study proposed a stepwise strategy based on the fused multisource remote sensing data to achieve fine-scale crop phenology monitoring across multistages, long-term spans, and large spatial scales. First, we utilized the STEPSBI method to fuse MODIS and Landsat, generating multispectral reflectance images covering six phenological stages with a spatial resolution of 30 m and a temporal resolution of eight days. Subsequently, input variables were screened according to their probability density distributions across different stages. Finally, phenology monitoring models for VGP and RGP were established using RF regression. We generated annual phenological datasets from 2001 to 2020 and analyzed the spatiotemporal variation characteristics of different phenological stages.

The stepwise strategy outperformed 1-D U-Net and SMF-S approaches, with a total RMSE of 8.46 days, by combining band reflectance, vegetation indices, and temperature-derived factors. It is reduced 15.7, 20.33, and 23.13 days compared with models using only one of these as inputs. This demonstrated better agreement with ground phenological observations. Furthermore, cross-validation indicated that the RMSEs of our dataset compared with Landsat (30 m), MODIS (500 m), and GLASS (1 km) were 4.37, 4.44, and 5.27 days, respectively. These results show consistency with existing phenological products. Our dataset better captures spatial heterogeneity and interannual variation at the field level. Moreover, the fusion accuracy of different spectral bands ranged from 0.812 to 0.875, demonstrating the reliability of the generated high-quality dense time-series imagery and effectively addressing the spatiotemporal resolution limitations of remote sensing data.

This study monitored the fine-scale phenology dynamic change across multiple stages, which are closely related to yield estimation, agricultural management, ecological restoration, and climate adaptation strategies. We identified various stages throughout the continuous crop growth cycle, overcoming the oversimplification in the conventional "start–end" phenology framework. The detailed phenological information improved the accuracy of crop classification and yield estimation, which is closely linked to optimizing crop planting structure. Furthermore, phenology datasets with 30-m resolution facilitate the identification of field-level phenological variations, reducing information loss associated with the regional average of coarse resolution pixels. This supports refined crop variety selection and effective management strategies. Additionally, long-term phenological information is essential for understanding photosynthesis, the carbon-water cycle, and energy flux dynamics within agricultural ecosystems, thereby promoting sustainable agricultural development.

Credit Authorship Contribution Statement

Yuyang Ma: conceptualization, formal analysis, investigation, methodology, software, validation, and writing—original draft. Haixiang Guan: supervision, resources, and writing—review and editing. Zhou Fang: formal analysis. Xuecao Li: resources and writing—review and editing. Jianxi Huang: conceptualization, funding acquisition, supervision, resources, and writing—review and editing.

References

[1] R. Cao et al., "A spatiotemporal shape model fitting method for within-season crop phenology detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 217, pp. 179–198, Nov. 2024.

[2] Y. Ma, Y. Shen, H. Guan, J. Wang, and C. Hu, "A novel approach to detect the spring corn phenology using layered strategy," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 122, Aug. 2023, Art. no. 103422.

[3] J. Pei et al., "The role of phenology in crop yield prediction: Comparison of ground-based phenology and remotely sensed phenology," *Agricult. Forest Meteorol.*, vol. 361, Feb. 2025, Art. no. 110340.

[4] F. Liu, X. Wang, Y. Zhang, and Y. Shen, "Novel metrics for assessing vegetation phenology from hydrological processes: Comparison with photosynthetic phenology," *Agricult. Forest Meteorol.*, vol. 358, Nov. 2024, Art. no. 110245.

[5] Y. Zeng et al., "Structural complexity biases vegetation greenness measures," *Nature Ecol. Evol.*, vol. 7, no. 11, pp. 1790–1798, Sep. 2023.

[6] Z. Zhu et al., "Greening of the Earth and its drivers," *Nat. Clim. Change*, vol. 6, no. 8, pp. 791–795, 2016.

[7] L. Zeng, B. D. Wardlow, D. Xiang, S. Hu, and D. Li, "A review of vegetation phenological metrics extraction using time-series, multispectral satellite data," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111511.

[8] Y. Pan, L. Li, J. Zhang, S. Liang, X. Zhu, and D. Sulla-Menashe, "Winter wheat area estimation from MODIS-EVI time series data using the crop proportion phenology index," *Remote Sens. Environ.*, vol. 119, pp. 232–242, Apr. 2012.

[9] L. Zeng et al., "A hybrid approach for detecting corn and soybean phenology with time-series MODIS data," *Remote Sens. Environ.*, vol. 181, pp. 237–250, Aug. 2016.

[10] B. Duchemin, J. Goubier, and G. Courrier, "Monitoring phenological key stages and cycle duration of temperate deciduous forest ecosystems with NOAA/AVHRR data," *Remote Sens. Environ.*, vol. 67, no. 1, pp. 68–82, Jan. 1999.

[11] A. Mercier et al., "Evaluation of Sentinel-1 & 2 time series for predicting wheat and rapeseed phenological stages," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 231–256, May 2020.

[12] Q. Niu et al., "A 30 m annual maize phenology dataset from 1985 to 2020 in China," *Earth Syst. Sci. Data*, vol. 14, no. 6, pp. 2851–2864, Jun. 2022.

[13] Y. Chen, R. Cao, J. Chen, L. Liu, and B. Matsushita, "A practical approach to reconstruct high-quality Landsat NDVI time-series data by gap filling and the Savitzky–Golay filter," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 174–190, Oct. 2021.

[14] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006.

[15] H. Gao et al., "CuFSDAF: An enhanced flexible spatiotemporal data fusion algorithm parallelized using graphics processing units," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.

[16] K. Knauer, U. Gessner, R. Fensholt, and C. Kuenzer, "An ESTARFM fusion framework for the generation of large-scale time series in cloud-prone and heterogeneous landscapes," *Remote Sens.*, vol. 8, no. 5, p. 425, May 2016.

[17] S. Liang et al., "Advancements in high-resolution land surface satellite products: A comprehensive review of inversion algorithms, products and challenges," *Sci. Remote Sens.*, vol. 10, Dec. 2024, Art. no. 100152.

[18] M. Liu et al., "An improved flexible spatiotemporal DAta fusion (IFSDAF) method for producing high spatiotemporal resolution normalized difference vegetation index time series," *Remote Sens. Environ.*, vol. 227, pp. 74–89, Jun. 2019.

[19] Y. Ma et al., "STEPSBI: Quick spatiotemporal fusion with coarse- and fine-resolution scale transformation errors and pixel-based synthesis base image pair," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 1–15, Dec. 2023.

[20] X. Zhang, S. Li, Z. Tan, and X. Li, "Enhanced wavelet based spatiotemporal fusion networks using cross-paired remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 281–297, May 2024.

[21] T. C. Nietupski, R. E. Kennedy, H. Temesgen, and B. K. Kerns, "Spatiotemporal image fusion in Google Earth engine for annual estimates of land surface phenology in a heterogenous landscape," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 99, Jul. 2021, Art. no. 102323.

[22] B. Sisheber, M. Marshall, D. Mengistu, and A. Nelson, "Tracking crop phenology in a highly dynamic landscape with knowledge-based Landsat–MODIS data fusion," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102670.

[23] B. Sisheber, M. Marshall, D. Mengistu, and A. Nelson, "Detecting the long-term spatiotemporal crop phenology changes in a highly fragmented agricultural landscape," *Agricult. Forest Meteorol.*, vol. 340, Sep. 2023, Art. no. 109601.

[24] X. Wang, S. Li, L. Wang, M. Zheng, Z. Wang, and K. Song, "Effects of cropland reclamation on soil organic carbon in China's black soil region over the past 35 years," *Global Change Biol.*, vol. 29, no. 18, pp. 5460–5477, Sep. 2023.

[25] L. Zhang, Z. Zhang, Y. Luo, J. Cao, and Z. Li, "Optimizing genotype-environment-management interactions for maize farmers to adapt to climate change in different agro-ecological zones across China," *Sci. Total Environ.*, vol. 728, Aug. 2020, Art. no. 138614.

[26] Y. Zhang et al., "Maize breeding for smaller tassels threatens yield under a warming climate," *Nature Climate Change*, vol. 14, no. 12, pp. 1306–1313, Dec. 2024.

[27] Y. Luo, Z. Zhang, Y. Chen, Z. Li, and F. Tao, "ChinaCropPhen1km: A high-resolution crop phenological dataset for three staple crops in China during 2000–2015 based on leaf area index (LAI) products," *Earth Syst. Sci. Data*, vol. 12, no. 1, pp. 197–214, Jan. 2020.

[28] J. G. Masek et al., "A Landsat surface reflectance dataset for North America, 1990–2000," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 68–72, Jan. 2006.

[29] Z. Zhu and C. E. Woodcock, "Object-based cloud and cloud shadow detection in Landsat imagery," *Remote Sens. Environ.*, vol. 118, pp. 83–94, Mar. 2012.

[30] N. Mishra, D. Helder, J. Barsi, and B. Markham, "Continuous calibration improvement: Landsat 5 through Landsat 8," *Remote Sens. Environ.*, vol. 185, pp. 7–15, Jun. 2016.

[31] A. S. Abowarda et al., "Generating surface soil moisture at 30 m spatial resolution using both data fusion and machine learning toward better water resources management at the field scale," *Remote Sens. Environ.*, vol. 255, Mar. 2021, Art. no. 112301, doi: 10.1016/j.rse.2021.112301.

[32] J. Rouse, R. H. Haas, J. Schell, and D. W. Deering, "Monitoring vegetation systems in the great plains with erts," *NASA Spec. Publ.*, vol. 1, pp. 309–317, Apr. 1973.

[33] A. R. Huete, "A soil-adjusted vegetation index (SAVI)," *Remote Sens. Environ.*, vol. 25, no. 3, pp. 295–309, Aug. 1988.

[34] J. Xue and B. Su, "Significant remote sensing vegetation indices: A review of developments and applications," *J. Sensors*, vol. 2017, pp. 1–17, Jun. 2017.

[35] Y. Zeng et al., "Optical vegetation indices for monitoring terrestrial ecosystems globally," *Nature Rev. Earth Environ.*, vol. 3, no. 7, pp. 477–493, May 2022.

[36] G. Rondeaux, M. Steven, and F. Baret, "Optimization of soil-adjusted vegetation indices," *Remote Sens. Environ.*, vol. 55, no. 2, pp. 95–107, Feb. 1996.

[37] A. Gitelson and M. N. Merzlyak, "Quantitative estimation of chlorophyll—A using reflectance spectra: Experiments with autumn chestnut and Maple leaves," *J. Photochem. Photobiol. B, Biol.*, vol. 22, no. 3, pp. 247–252, Mar. 1994.

[38] A. A. Gitelson, "Wide dynamic range vegetation index for remote quantification of biophysical characteristics of vegetation," *J. Plant Physiol.*, vol. 161, no. 2, pp. 165–173, 2004.

[39] R. Taghizadeh-Mehrjardi, K. Nabiollahi, and R. Kerry, "Digital mapping of soil organic carbon at multiple depths using different data mining techniques in Baneh region, Iran," *Geoderma*, vol. 266, pp. 98–110, Mar. 2016.

[40] W. Yang et al., "MODIS leaf area index products: From validation to algorithm improvement," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 7, pp. 1885–1898, Jul. 2006.

[41] C. Liao, J. Wang, B. Shan, J. Shang, T. Dong, and Y. He, "Near real-time detection and forecasting of within-field phenology of winter wheat and corn using Sentinel-2 time-series data," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 105–119, Feb. 2023.

[42] D. C. Lafferty, R. L. Sriver, I. Haqiqi, T. W. Hertel, K. Keller, and R. E. Nicholas, "Statistically bias-corrected and downscaled climate models underestimate the adverse effects of extreme heat on U.S. maize yields," *Commun. Earth Environ.*, vol. 2, no. 1, Sep. 2021, Art. no. 196.

[43] Q. Peng et al., "A twenty-year dataset of high-resolution maize distribution in China," *Scientific Data*, vol. 10, no. 1, Sep. 2023, Art. no. 658.

[44] L. Liu et al., "Detecting crop phenology from vegetation index time-series data by improved shape model fitting in each phenological stage," *Remote Sens. Environ.*, vol. 277, Aug. 2022, Art. no. 113060.

[45] F. Lobert et al., "A deep learning approach for deriving winter wheat phenology from optical and SAR time series at field level," *Remote Sens. Environ.*, vol. 298, Dec. 2023, Art. no. 113800.

[46] Q. Tian et al., "PSeqNet: A crop phenology monitoring model accounting for phenological associations," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 257–274, Jul. 2025.

[47] J. Gray, D. Sulla-Menashe, and M. A. Friedl, "User guide to collection 6 MODIS land cover dynamics (MCD12Q2) product," NASA EOSDIS Land Processes DAAC, Sioux Falls, SD, USA, Tech. Rep., 2019, pp. 1–8, vol. 6.

[48] L. V. D. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.

[49] Y. Wen et al., "Mapping corn dynamics using limited but representative samples with adaptive strategies," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 252–266, Aug. 2022.

[50] J. Chen, Y. Liu, W. Zhou, J. Zhang, and T. Pan, "Effects of climate change and crop management on changes in Rice phenology in China from 1981 to 2010," *J. Sci. Food Agricult.*, vol. 101, no. 15, pp. 6311–6319, Dec. 2021.

[51] Z. Peng et al., "Impacts of the scale effect on quantifying the response of spring vegetation phenology to urban intensity," *Remote Sens. Environ.*, vol. 315, Dec. 2024, Art. no. 114485.

[52] S. Testa, K. Soudani, L. Boschetti, and E. Borgogno Mondino, "MODIS-derived EVI, NDVI and WDRVI time series to estimate phenological metrics in French deciduous forests," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 64, pp. 132–144, Feb. 2018.

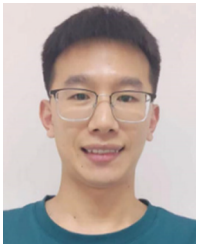
[53] T. Sakamoto, B. D. Wardlow, A. A. Gitelson, S. B. Verma, A. E. Suyker, and T. J. Arkebauer, "A two-step filtering approach for detecting maize and soybean phenology with time-series MODIS data," *Remote Sens. Environ.*, vol. 114, no. 10, pp. 2146–2159, Oct. 2010.

[54] T. H. Ramirez-Parada, I. W. Park, S. Record, C. C. Davis, A. M. Ellison, and S. J. Mazer, "Plasticity and not adaptation is the primary source of temperature-mediated variation in flowering phenology in North America," *Nature Ecology Evol.*, vol. 8, no. 3, pp. 467–476, Jan. 2024.

[55] M. Zhou et al., "Detection of phenology using an improved shape model on time-series vegetation index in wheat," *Comput. Electron. Agric.*, vol. 173, Jul. 2020, Art. no. 105398.
[56] M. Wiesmeier et al., "Soil organic carbon storage as a key function of soils—A review of drivers and indicators at various scales," *Geoderma*, vol. 333, pp. 149–162, Jan. 2019.
[57] H. Qiu et al., "Flowering in the northern Hemisphere is delayed by frost after leaf-out," *Nature Commun.*, vol. 15, no. 1, p. 9123, 2024.
[58] R. Chen et al., "Flowering delay in apple could alleviate frost-induced yield loss under climate change in China," *Agricult. Forest Meteorol.*, vol. 360, Jan. 2025, Art. no. 110313.
[59] Z. Gong, W. Ge, J. Guo, and J. Liu, "Satellite remote sensing of vegetation phenology: Progress, challenges, and opportunities," *ISPRS J. Photogramm. Remote Sens.*, vol. 217, pp. 149–164, Nov. 2024.
[60] Z. Li, Z. Zhang, J. Zhang, Y. Luo, and L. Zhang, "A new framework to quantify maize production risk from chilling injury in Northeast China," *Clim. Risk Manag.*, vol. 32, Jan. 2021, Art. no. 100299.

Yuyang Ma received the bachelor's and master's degrees from the School of Resources and Environment and the School of Public Administration and Law, Northeast Agricultural University, Harbin, China, in 2018 and 2021, respectively, and the Ph.D. degree from the School of Geography and Information Engineering, China University of Geosciences (Wuhan), Wuhan, China, in 2025.

She is currently a Lecturer with the College of Geography and Planning, Chengdu University of Technology, Chengdu, China. Her research focuses on crop phenology stage recognition, deep learning, and spatiotemporal fusion.



Haixiang Guan received the bachelor's degree from the School of Law and Public Administration, Huaihua University, Huaihua, China, in 2018, the master's degree from the School of Public Administration and Law, Northeast Agricultural University, Harbin, China, in 2021, and the Ph.D. degree from the College of Land Science and Technology, China Agricultural University, Beijing, China, in 2025.

He is currently an Associate Professor with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu, China. His research focuses on agricultural disaster monitoring and machine learning with their applications in remote sensing.

Zhou Fang received the bachelor's degree from the School of Geosciences and Engineering, Southwest Jiatong University, Chengdu, China, in 2024, where he is currently pursuing the master's degree.

His research interests include remote sensing, deep learning, and crop phenology stage recognition.

Yan Shuai received the B.Eng. degree in surveying engineering from Shandong University of Technology, Zibo, Shandong, China, in 2017, the M.Sc. degree in surveying engineering from China University of Geosciences (Beijing), Beijing, China, in 2020, and the Ph.D. degree in land resource management from China Agricultural University, Beijing, in 2024.

He currently serves as an Assistant Researcher with Shandong Rural Remote Sensing Application Center, Shandong Academy of Agricultural Sciences, Jinan, Shandong. His work has led to advancements in land environment (i.e., sea/lake ice, land use, and land cover) mapping. He has published various articles in high-impact remote sensing journals, such as the *International Journal of Applied Earth Observation and Geoinformation* and *Computers and Electronics in Agriculture*. His research is at the cutting edge of integrating machine learning, AI, and remote sensing technologies, including SAR and multispectral, to tackle pressing environmental challenges.

Bo Huang received the B.Sc. degree in geographic information science from Shandong Agricultural University, Tai'an, China, in 2024. He is currently pursuing the master's degree with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu, China.

His research interests include multisource data processing in agriculture and crop phenology monitoring.

Xuecao Li received the B.Sc. degree in geographic information science from Sun Yat-sen University, Guangzhou, China, in 2012, and the Ph.D. degree in ecology from Tsinghua University, Beijing, China, in 2016.

He is currently a Professor with the College of Land Science and Technology, China Agricultural University, Beijing. He has authored more than 90 articles in international journals. His research interests include land use/cover change and modeling, global environmental change, and sustainable development.

Jianxi Huang (Senior Member, IEEE) received the B.Sc. degree in cartography and geographic information system from Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1999, the M.Sc. degree in cartography and geographic information system from Wuhan University, Wuhan, in 2002, and the Ph.D. degree in agricultural remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2006.

He is currently a Professor with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu, China. He has authored more than 60 articles in international journals. His research interests include data assimilation applications in agriculture, crop yield estimation and forecasting, and agricultural drought monitoring and assessment.